

Observer-Based Sampled-Data Stabilisation of Switched Systems with Lipschitz Nonlinearities and Dwell-Time^{*}

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Abstract: We investigate the stabilisation of nominally linear-affine switched systems with uncertain Lipschitz nonlinearities under dwell-time constraints, using a sampled-data switching law based on a state observer. We design the switching law based on Lyapunov-Metzler inequalities, accounting for the sampled-data output measurements, and we derive time-dependent LMI conditions for global asymptotic stability (or, in the presence of switching affine terms, ultimate boundedness) of the resulting closed-loop system. We obtain an estimate of the average quadratic cost and a bound on its maximum deviation from the actual cost. Moreover, we discuss the feasibility of the derived LMIs. Specifically, we show how the observer gains can be incorporated into the matrix inequalities, provide equivalent reduced-order LMI conditions, and prove that the time dependence of the LMIs can be removed by discretising on a finite grid. Numerical examples, including practical applications to real-world engineering scenarios in power systems, illustrate our theoretical results and compare them with an existing approach for output-feedback stabilisation of switched systems, subject to sampled-data measurements.

Keywords: Switched systems, uncertain systems, observers, sampled-data, dwell time, Lyapunov-Metzler inequalities.

1. INTRODUCTION

Switched systems are composed of multiple modes, each governed by a distinct dynamical system, while a switching law dictates the transitions between modes (Liberzon, 2003). The switched system does not necessarily retain the stability characteristics of the individual modes; hence, it is crucial to analyse the stability of the switched system under various switching conditions (Liberzon and Morse, 1999) and to design stabilising switching strategies (Geromel and Colaneri, 2006b). Classic design approaches rely either on arbitrarily fast switching among modes or on a dwell time, i.e., a minimum amount of time between consecutive switching instants (Geromel and Colaneri, 2006b). Particular attention has been devoted

to switched systems that are either linear or affine, in continuous time (Johansson and Rantzer, 1998; Hespanha, 2004; Bolzern and Spinelli, 2004; Geromel and Colaneri, 2006b; Albea and Seuret, 2021; Russo et al., 2025) and in discrete time (Daafouz and Bernussou, 2001; Geromel and Colaneri, 2006a; Egidio and Deaecto, 2019, 2021). While the literature mostly considers full state feedback, a body of work focuses on the design of stabilising output feedback laws (Faron, 1996; Decarlo et al., 2000; Deaecto, 2016; Deaecto and Daiha, 2020; Deaecto et al., 2023); e.g., Geromel et al. (2008) provide conditions to stabilise continuous- and discrete-time switched systems through a feedback controller based on output measurements, relying on Lyapunov-Metzler inequalities. As an additional challenge, the output may be only available at discrete sampling instants, which requires sampled-data control methods, explored in the context of switched systems, e.g., by Hetel and Fridman (2013); Albea et al. (2024).

We consider the stabilisation of a class of nominally linear switched systems, possibly including an affine switching term, affected by uncertain Lipschitz nonlinearities, under dwell-time constraints, sampled-data measurements, and

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output feedback (Section 2). In Section 3, we propose a novel switching law that relies on a state observer, along with Lyapunov-Metzler inequalities (Russo et al., 2025) accounting for dwell-time constraints. Specifically, we discuss the observer design in Section 3.1 and the design of the observer gains in Section 3.3. In Section 3.2 we introduce our proposed observer-based sampled-data switching law, subject to dwell-time constraints, and we derive time-dependent LMI conditions ensuring global asymptotic stability (or, in the presence of switching affine terms, ultimate boundedness) of the closed-loop system. We also obtain an estimate of the average quadratic cost, together with a bound on its maximum deviation from the actual cost function. Moreover, in the presence of switching affine terms with a known common upper bound, we establish explicit ultimate bounds. In the case of known affine terms, Section 3.4 presents a modified observer-based switching-law design, aimed at improving the estimates on the obtained ultimate bound for the system. In Section 4, we discuss the feasibility of the derived LMIs, providing equivalent reduced-order LMI conditions, and prove that the time dependence of the LMIs can be removed by discretisation on a finite grid. Numerical examples in Section 5 illustrate the efficacy of our proposed method, even at stabilising switched systems with unstable modes and with modes having no stable convex combination, concurrently addressing output feedback, sampled-data and dwell-time constraints. The possible presence of affine terms in the dynamics allows us to address real-life engineering applications, such as power converters (Russo et al., 2022; Carneiro et al., 2024), where affine terms usually model the presence of an energy source acting on the system: in particular, we consider the realistic example of a double buck-boost converter, which we use to showcase our approach and compare it with the approach proposed by Hetel and Fridman (2013), which deals with sampled-data state-feedback stabilisation, but does not include filter design and/or Lyapunov-Metzler inequalities. In the work by Hetel and Fridman (2013), the dwell time is equal to the inter-sampling time of the state, and hence dwell-time constraints cannot be accommodated into the switching signal. The numerical results show that our methodology performs at least as well under the same conditions and can actually perform better, due to the inherent decoupling between the sampling sequence in the estimation channel and the switching signal updates. To the best of our knowledge, no existing contribution integrates all the features we consider within a unified observer-based sampled-data framework. Other works in the literature, related to specific aspects of our problem formulation, are discussed in detail in Section 6, which also draws conclusions and discusses future work.

Notation. Given matrix A , we denote by $A^\top$ its transpose and by A_{ij} its (i, j) -th element. When A is symmetric, its entries below the sub-diagonal are denoted by $\star$. $\mathbb{R}_{\geq 0}$ and $\mathbb{N}$ are the sets of non-negative real numbers and natural numbers, respectively. We denote $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$. For $\ell \in \mathbb{N}$, $[\ell]$ denotes the set $\{1, \dots, \ell\}$, whereas $\mathcal{M}_\ell \subseteq \mathbb{R}^{\ell \times \ell}$ denotes the class of irreducible Metzler matrices such that for all matrices $\Pi \in \mathcal{M}_\ell$, it holds that $\Pi_{ij} \geq 0$ for all $i \neq j$ and $\sum_{j=1}^\ell \Pi_{ij} = 0$ for all $i \in [\ell]$. For a symmetric matrix $X \in \mathbb{R}^{n \times n}$, $X \prec 0$ (respectively, $X \succ 0$) indicates

that X is positive (respectively, negative) definite. Finally, I_n denotes the n -dimensional identity matrix and $\|\cdot\|$ the Euclidean norm. The corresponding induced operator norm on $\mathbb{R}^{n \times n}$ is also denoted by $\|\cdot\|$. For matrices $\{A_i\}_{i=1}^k$ of appropriate size, $\text{col}\{A_i\}_{i=1}^k$ is the block matrix stacking the A_i 's in consecutive rows.

2. PROBLEM FORMULATION

Given the state $x \in \mathbb{R}^n$, measured output $y \in \mathbb{R}^m$ and performance output $z \in \mathbb{R}^p$, consider the switched system

$$\begin{aligned} \dot{x}(t) &= A_{\sigma(t)}x(t) + \theta b_{\sigma(t)} + f_{\sigma(t)}(x(t)), & x(t_0) &= x_0, \\ y(t) &= D_{\sigma(t)}x(s_k), & t &\in [s_k, s_{k+1}), \\ z(t) &= Cx(t), \end{aligned} \quad (1)$$

with *unknown* initial condition $x(t_0)$. We aim to design the piecewise-constant switching law $\sigma: \mathbb{R}_{\geq 0} \rightarrow [\ell]$, $\ell \in \mathbb{N}$, subject to dwell-time constraints. Thus, denoting with $\mathcal{T} = \{t_k\}_{k=0}^\infty$ the increasing sequence of switching instants, we introduce a dwell time $T > 0$ such that $t_{k+1} - t_k \geq T$ for all $k \in \mathbb{N}$. Moreover, the output is sampled at sampling instants $\mathcal{S} = \{s_k\}_{k=0}^\infty$. Our results allow for the values of the sampling instants $\mathcal{S}$ to be arbitrary, provided that $0 < s_{k+1} - s_k \leq h$ for all $k \in \mathbb{N}$, where $h > 0$ is a maximum inter-sampling time, and $\lim_{k \rightarrow \infty} s_k = \infty$. The bound h will be explicitly obtained from suitable matrix inequalities (see e.g., Theorem 1 in Section 3.2). Henceforth we will assume, without loss of generality, that $t_0 = s_0 = 0$. Finally, the matrices $A_{\sigma(t)} \in \{A_j\}_{j \in [\ell]}$, $D_{\sigma(t)} \in \{D_j\}_{j \in [\ell]}$ and C are *known*, whereas the vectors $b_{\sigma(t)} \in \{b_j\}_{j \in [\ell]}$ are *unknown*; however, an upper bound $\bar{b}$ on their norms is available, meaning that

$$\max_{i \in [\ell]} \|b_i\| \leq \bar{b}. \quad (2)$$

Similarly, the functions $f_{\sigma(t)} \in \{f_j\}_{j \in [\ell]}$, with $f_j: \mathbb{R}^n \rightarrow \mathbb{R}^n$, $j \in [\ell]$, are *unknown*, and represent unknown and unmodelled state-dependent dynamics. The parameter θ in (1) satisfies $\theta \in \{0, 1\}$, where for $\theta = 0$ we will examine the stability of the obtained closed-loop system, whereas for $\theta = 1$ we will study its ultimate boundedness.

We make the following standing assumptions.

Assumption 1. Functions $\{f_i\}_{i \in [\ell]}$ are globally Lipschitz, i.e., there exist $\kappa > 0$, such that, for all $x, y \in \mathbb{R}^n$,

$$\|f_i(x) - f_i(y)\| \leq \kappa \|x - y\|, \quad \forall i \in [\ell]. \quad (3)$$

Moreover, for all $i \in [\ell]$, $f_i(0) = 0$. The generality of this assumption is discussed in Remarks 3 and 8.

Remark 1. The subsequent analysis shows that the upper bound κ in (3) can be replaced by mode-dependent constants $\kappa_i > 0$, $i \in [\ell]$. Whenever such tighter mode-dependent bounds are available, the approach presented in this manuscript is expected to yield less conservative results.

Assumption 2. For all $i \in [\ell]$, the pair (A_i, D_i) is observable. Moreover, there exist $\{L_i\}_{i \in [\ell]}$ such that for all $i \in [\ell]$

$$U_i := A_i - L_i D_i \quad (4)$$

is Hurwitz and the matrices $\{U_i\}_{i \in [\ell]}$ are simultaneously quadratically Lyapunov stable, i.e., there exist $\Omega \in \mathbb{R}^{n \times n}$, with $\Omega \succ 0$, and $\eta > 0$ such that

$$\Phi_i(\eta, \Omega) := U_i^\top \Omega + \Omega U_i + 2\eta \Omega \prec 0, \quad \forall i \in [\ell]. \quad (5)$$

We further define the average quadratic cost functional

$$J(t) = \frac{1}{t} \int_0^t z(\tau)^\top z(\tau) d\tau, \quad t > 0, \quad (6)$$

which accounts for the average energy of the performance output z . The value of J may be considered as a measure of the efficiency of the proposed switching algorithm and is of great importance when selecting the best approach in applications, as it serves as a robustness indicator of the process behaviour. Under our assumptions $x(t)$, $t \geq 0$, is not observed, whence J cannot be computed directly.

Considering the system (1) and the presented standing assumptions, we now formulate our control objectives.

Objective 1. Using the available sampled-data measurements, design an observer $\varphi(t)$ and an observer-based switching law $\sigma(\varphi(t), t): \mathbb{R}^m \times \mathbb{R} \rightarrow [\ell]$ such that the closed-loop system $(x(t), \varphi(t))$ is globally asymptotically stable ($\theta = 0$) or ultimately bounded ($\theta = 1$). In both cases, we require explicit bounds on the decay rate of the closed-loop system and the maximum inter-sampling time $h > 0$. For $\theta = 1$, we further require a computable upper estimate on the ultimate bound of the state $(x(t), \varphi(t))$.

Objective 2. Let $\theta = 0$ and assume that an a-priori norm estimate $\|x(0)\| \leq R$, with $R > 0$, on the unknown initial condition is available. Provide a computable estimate $\hat{J}(t)$ of the average cost (6) and derive an explicit error bound on $|J(t) - \hat{J}(t)|$, which holds pointwise for all $t \geq 0$. Given $\varepsilon > 0$, compute a time $t_* = t_*(\varepsilon, R) > 0$ such that $t \geq t_*$ implies $|J(t) - \hat{J}(t)| < \varepsilon$. For $\theta = 1$, provide a computable ultimate bound on the discrepancy $|J(t) - \hat{J}(t)|$ between the actual average cost functional $J(t)$ and the provided estimate $\hat{J}(t)$.

Remark 2. Assumption 2 has been investigated in the literature in the context of polytopic-type uncertainty for parameter-dependent systems of the form $\dot{w} = F(\lambda)w$, where λ is a parameter (see, e.g., Boyd and Yang, 1989). Since $\{A_i\}_{i \in [\ell]}$ and $\{D_i\}_{i \in [\ell]}$ are known, Assumption 2 can be verified computationally by solving

$$\begin{cases} \Omega A_i - Y_i D_i + A_i^\top \Omega - D_i^\top Y_i^\top + 2\eta \Omega \prec 0, & i \in [\ell], \\ \Omega \succ 0 \end{cases}$$

for some $\Omega \succ 0$ and $\{Y_i\}_{i \in [\ell]}$ of appropriate dimensions. If the above inequalities are feasible, $\{L_i\}_{i \in [\ell]}$ can be set as

$$L_i = \Omega^{-1} Y_i, \quad i \in [\ell]. \quad (7)$$

By (Boyd et al., 1994) and Finsler's lemma (Schweppe, 1973), Assumption 2 holds iff there exist some $\Omega \succ 0$, $\eta > 0$ and $\varpi_i \in \mathbb{R}$, $i \in [\ell]$, such that for each $i \in [\ell]$

$$(A_i + \eta I_n)^\top \Omega + \Omega (A_i + \eta I_n) x \prec \varpi_i D_i^\top D_i,$$

iff the systems $\dot{z}_i(t) = (A_i + \eta I_n)z_i(t)$, $v_i(t) = D_i z_i(t)$, $i \in [\ell]$ are jointly dissipative with a common storage function $S(z) = z^\top \Omega z$ and supply rates $\omega_i(v_i) = \varpi_i v_i^\top v_i$ (Van der Schaft, 2000). Section 3.3 further discusses the design of the observer gains $\{L_i\}_{i \in [\ell]}$.

Remark 3. Our proposed approach is also suitable for the regional stabilisation of systems of the form (1), where $\theta = 0$ and the nonlinearities $\{f_i\}_{i \in [\ell]}$ satisfy $f_i(0) = 0$, as well as a local Lipschitz condition $\|f_i(x) - f_i(y)\| \leq \kappa(\rho) \|x - y\|$, for $\|x\|, \|y\| \leq \rho$, with known local Lipschitz constant $\kappa(\rho)$ satisfying

$$\lim_{\rho \rightarrow 0^+} \kappa(\rho) = 0. \quad (8)$$

Condition (8) guarantees that, in a neighbourhood of the origin, the linear parts of the dynamics in (1) with $\theta = 0$ dominate the nonlinearities $\{f_i\}_{i \in [\ell]}$, thereby making our approach suitable for local stabilisation of (1). Indeed, one can show that subject to (8), Theorem 1 below and its proof can be applied locally around the origin. When mode-dependent Lipschitz constants $\kappa_i(\rho)$ are available (see Remark 1), condition (8) becomes $\lim_{\rho \rightarrow 0^+} \max_{i \in [\ell]} \kappa_i(\rho) = 0$. For concreteness of the presentation, we proceed with the case where $\{f_i\}_{i \in [\ell]}$ are globally Lipschitz.

Previously proposed switching strategies such as those by Albea et al. (2019); Albea and Seuret (2021); Russo et al. (2025) cannot be employed to control system (1), since both $x(0)$ and the right-hand side of (1) are unknown.

3. OBSERVER-BASED SWITCHING LAW DESIGN

To achieve the control Objectives 1 and 2, we propose an observer-based sampled-data switching strategy, subject to dwell-time constraints. The observer is designed to compensate the unavailability of $x(t)$, $t \geq 0$ for the design of the switching strategy. For simplicity, we denote the observer-based switching strategy $\sigma(\varphi(t), t)$ as $\sigma(t)$.

3.1 Observer design

Given any switching law $\sigma(t)$ for system (1), we consider the observer $\varphi: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$ satisfying

$$\begin{aligned} \dot{\varphi}(t) &= A_{\sigma(t)} \varphi(t) + L_{\sigma(t)} [y(s_k) - D_{\sigma(t)} \varphi(s_k)], \\ t &\in [s_k, s_{k+1}), \\ \varphi(0) &= 0, \end{aligned} \quad (9)$$

where $\{L_i\}_{i \in [\ell]}$ are observer gains satisfying Assumption 2. Observer (9) relies solely on knowledge of $\sigma(t)$, $t \geq 0$ (to be designed shortly), of $A_{\sigma(t)}$ and $D_{\sigma(t)}$, and of the sampled output $y(s_k) = D_{\sigma(t)} x(s_k)$, $t \in [s_k, s_{k+1})$, all of which are assumed to be known. In particular, b_i , $i \in [\ell]$, and f_i , $i \in [\ell]$, which are assumed unknown, do not appear in the observer design (9).

Denote the observer error as $e(t) := x(t) - \varphi(t)$. By (1) and (9), we obtain

$$\begin{aligned} \dot{e}(t) &= A_{\sigma(t)} e(t) - L_{\sigma(t)} D_{\sigma(t)} e(s_k) + \theta b_{\sigma(t)} + f_{\sigma(t)}(x(t)), \\ t &\in [s_k, s_{k+1}), \\ e(0) &= x(0). \end{aligned}$$

Denoting by $\delta_e(t) := e(t) - e(s_k)$, $t \in [s_k, s_{k+1})$, we obtain

$$\begin{aligned} \dot{e}(t) &= [A_{\sigma(t)} - L_{\sigma(t)} D_{\sigma(t)}] e(t) + L_{\sigma(t)} D_{\sigma(t)} \delta_e(t) \\ &\quad + \theta b_{\sigma(t)} + f_{\sigma(t)}(x(t)) \\ &= U_{\sigma(t)} e(t) + L_{\sigma(t)} D_{\sigma(t)} \delta_e(t) + \theta b_{\sigma(t)} + f_{\sigma(t)}(x(t)), \\ t &\in [s_k, s_{k+1}). \end{aligned} \quad (10)$$

Since $\lim_{k \rightarrow \infty} s_k = \lim_{k \rightarrow \infty} t_k = \infty$, solutions $\varphi(t)$ and $e(t)$ of (9) and (10), respectively, exist and are absolutely continuous, with the corresponding ODEs satisfied outside of a discrete set of instants $\mathfrak{T}$, such that $\mathfrak{T}$ has a finite intersection with every bounded set in $[0, \infty)$.

Finally, since $x(s_k) - \varphi(s_k) = e(s_k) = e(t) - \delta_e(t)$ and $x(t) = \varphi(t) + e(t)$, we present the closed-loop system in terms of $(\varphi(t), e(t))$ as follows

$$\begin{aligned}\dot{\varphi}(t) &= A_{\sigma(t)}\varphi(t) + L_{\sigma(t)}D_{\sigma(t)}e(t) - L_{\sigma(t)}D_{\sigma(t)}\delta_e(t) \\ \dot{e}(t) &= U_{\sigma(t)}e(t) + L_{\sigma(t)}D_{\sigma(t)}\delta_e(t) \\ &\quad + \theta b_{\sigma(t)} + f_{\sigma(t)}(\varphi(t) + e(t)), \quad t \geq 0.\end{aligned}\quad (11)$$

We now proceed to design the switching law $\sigma(t)$, $t \geq 0$, that achieves the control Objectives 1 and 2.

3.2 Proposed switching strategy

We first introduce auxiliary variables that we will use in formulating the switching law $\sigma(t)$, $t \geq 0$, and our main result. Consider a scalar $\zeta > 0$, matrices $\{X_i\}_{i \in [\ell]} \subseteq \mathbb{R}^{n \times n}$, $X_i \succ 0$, $i \in [\ell]$, and an irreducible Metzler matrix $\Pi \in \mathcal{M}_\ell$ that satisfy the Lyapunov-Metzler inequalities

$$A_i^\top X_i + X_i A_i + \sum_{j \in [\ell] \setminus \{i\}} \Pi_{ij} (Y_{1,j} + Y_{2,j} - X_i) + C^\top C + 2\zeta X_i \prec 0, \quad (12)$$

where, for each $j \in [\ell]$,

$$\begin{aligned}Y_{1,j} &= e^{(A_j^\top + \zeta I)T} X_j e^{(A_j + \zeta I)T}, \\ Y_{2,j} &= \int_0^T e^{(A_j^\top + \zeta I)\tau} C^\top C e^{(A_j + \zeta I)\tau} d\tau.\end{aligned}\quad (13)$$

We introduce an observer-based switching law, with dwell-time $T > 0$, as follows. Given $t_k \in [0, \infty)$,

$$\sigma(t) = i, \quad t \in [t_k, t_k + T], \quad (14a)$$

$$\sigma(t) = i, \quad t > t_k + T, \quad \text{as long as, } \forall j \in [\ell] \setminus \{i\}, \quad (14b)$$

$$\text{it is } \varphi(t)^\top (Y_{1,j} + Y_{2,j}) \varphi(t) \geq \varphi(t)^\top X_i \varphi(t),$$

$$\sigma(t_{k+1}) = \underset{j \in [\ell] \setminus \{i\}}{\operatorname{argmin}} [\varphi(t_{k+1})^\top (Y_{1,j} + Y_{2,j}) \varphi(t_{k+1})], \quad (14c)$$

where

$$t_{k+1} := \inf_{t > t_k + T} \{t \mid \exists j: \varphi(t)^\top [Y_{1,j} + Y_{2,j} - X_i] \varphi(t) < 0\}.$$

Remark 4. As can be seen from (14), the set $\{t_k\}_{k=0}^\infty$ can exhibit one of two possible behaviours. First, there may be t_k such that for $t \geq t_k$, the condition in (14b) is never violated, and the system “stabilises” on a single operational mode: $\sigma(t) \equiv i$ for $t \geq t_k$. In this case, we formally define $t_{k+1} \equiv \infty$ for all $j \geq 1$, thereby guaranteeing that $\lim_{k \rightarrow \infty} t_k = \infty$, in the extended real line $\bar{\mathbb{R}} = \mathbb{R} \cup \{\infty\}$. Second, there may be infinitely many violations of the condition in (14b), thereby leading to $t_k < \infty$ for all k and, since $t_{k+1} - t_k \geq T > 0$, $\lim_{k \rightarrow \infty} t_k = \infty$ in $\mathbb{R}$. For our stability/ultimate boundedness results, as seen in, e.g., the proof of Theorem 1, the distinction between the two behaviours of $\{t_k\}_{k=0}^\infty$ is immaterial.

Remark 5. The switching law (14) requires knowledge of the matrices A_i , $i \in [\ell]$, but not of b_i , $i \in [\ell]$, and f_i , $i \in [\ell]$, which are assumed unknown. When b_i , $i \in [\ell]$, are known, incorporating them into both the observer and the switching law designs greatly improves the conservatism of the obtained ultimate bound: in Section 3.4, we modify our observer-based switching law accordingly.

We are now ready to state our main result.

Theorem 1. Consider the closed-loop system (11) under Assumptions 1 and 2, with gains L_i fixed (e.g., as in Remark 2). Let $\zeta, T > 0$, $\{X_i\}_{i \in [\ell]} \subseteq \mathbb{R}^{n \times n}$, $X_i \succ 0$, $i \in [\ell]$,

and $\Pi \in \mathcal{M}_\ell$ satisfy the Lyapunov-Metzler inequality (12), and consider the time-varying symmetric positive-definite matrices $P_i: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{n \times n}$ such that

$$-\dot{P}_i(t) = A_i^\top P_i(t) + P_i(t) A_i + C^\top C + 2\zeta P_i(t), \quad t \in [t_k, t_k + T], \quad i \in [\ell], \quad (15a)$$

$$P_i(t) = X_i, \quad t \in [t_k + T, t_{k+1}), \quad i \in [\ell]. \quad (15b)$$

Given tuning parameters $h > 0$, $\alpha > 0$ and $\kappa > 0$, $i \in [\ell]$, let there exist matrices $Q \succ 0$, $W \succ 0$ and scalar $\gamma > 0$ such that the LMIs $\Psi_i(t) \prec 0$, with $\{\Psi_i(t)\}_{i \in [\ell]}$ given in (16), hold for all $i \in [\ell]$ and all $t \in [0, T]$. Then,

- I. For $\theta = 0$ in (1), the observer-based switching law (14) yields global asymptotic stability of the origin for the closed-loop system (11), meaning that

$$\|\varphi(t)\|^2 + \|e(t)\|^2 \leq M e^{-2\alpha t} \|\varphi(0)\|^2, \quad t \geq 0, \quad (17)$$

for some $M \geq 1$, thereby satisfying Objective 1. The estimate $\hat{J}(t)$ of the average cost, given by

$$\hat{J}(t) = \frac{1}{t} \int_0^t \varphi(s)^\top C^\top C \varphi(s) ds, \quad t > 0, \quad (18)$$

satisfies Objective 2.

- II. For $\theta = 1$ in (1), the observer-based switching law (14) guarantees that the solutions of the closed-loop system (11) are ultimately bounded, meaning that

$$\begin{aligned}\limsup_{t \rightarrow \infty} \left(\|\varphi(t)\|^2 + \|e(t)\|^2 \right) &\leq \frac{\gamma \bar{b}^2}{2\alpha \min(\mathcal{C}_1, \mathcal{C}_3)} =: \bar{B}, \\ \limsup_{t \rightarrow \infty} \|x(t)\|^2 &\leq 2\bar{B},\end{aligned}\quad (19)$$

with explicitly computable $\bar{B}$. Moreover, the estimate $\hat{J}(t)$ of the average cost satisfies

$$\limsup_{t \rightarrow \infty} |J(t) - \hat{J}(t)| \leq 3 \|C^\top C\| \bar{B}^2. \quad (20)$$

Remark 6. Note that (15) is an *autonomous* system that is solved on each sub-interval $[t_k, t_k + T]$ with the *same* boundary value $P_i(t) = X_i$, for the fixed matrix X_i , whereas on $[t_k + T, t_{k+1})$ we have $P_i \equiv X_i$. Thus, it can be seen that $P_i: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{n \times n}$ is fully determined by its values on $[0, T]$ (recall that $t_0 = 0$ by assumption). Hence, $\Psi_i(t) \prec 0$ for $t \in [0, T]$ guarantees $\Psi_i(t) \prec 0$ for all $t \geq 0$. Therefore, in Theorem 1, $\Psi_i(t) \prec 0$ is required to hold only for $t \in [0, T]$, and not for all $t \in [t_k, t_k + T]$, $k \in \mathbb{N}_0$.

Proof. Let $Q, W \in \mathbb{R}^{n \times n}$ with $Q, W \succ 0$ and $\alpha \in (0, \zeta)$. Consider the candidate Lyapunov functional

$$V(t) = V_\varphi(t) + V_e(t) \quad (21)$$

where

$$V_\varphi(t) = \varphi(t)^\top P_{\sigma(t)}(t) \varphi(t), \quad t \geq 0 \quad (22)$$

and

$$\begin{aligned}V_e(t) &= e(t)^\top Q e(t) + h^2 e^{2\alpha h} \int_{s_k}^t e^{-2\alpha(t-s)} \dot{e}(s)^\top W \dot{e}(s) ds \\ &\quad - \frac{\pi^2}{4} \int_{s_k}^t e^{-2\alpha(t-s)} \delta_e(s)^\top W \delta_e(s) ds, \quad t \in [s_k, s_{k+1}).\end{aligned}\quad (23)$$

$V_\varphi(t)$ is continuous on each interval $[t_k, t_{k+1})$, $k \in \mathbb{N}_0$, differentiable on each interval (t_k, t_{k+1}) , $k \in \mathbb{N}_0$ and satisfies

$$\mathcal{C}_1 \|\varphi(t)\|^2 \leq V_\varphi(t) \leq \mathcal{C}_2 \|\varphi(t)\|^2, \quad t \geq 0, \quad (24)$$

for some explicitly computable $\mathcal{C}_i > 0$, $i = 1, 2$, in view of the positive-definiteness and continuity of $P_i(t)$,

$$\Psi_i(t) := \begin{bmatrix} -2(\zeta - \alpha)P_i(t) + \gamma\kappa^2 I_n & P_i(t)L_i D_i + \gamma\kappa^2 I_n & -P_i(t)L_i D_i & 0 \\ * & \Phi_i(\alpha, Q) + \gamma\kappa^2 I_n + h^2 e^{2\alpha h} U_i^\top W U_i & Q L_i D_i + h^2 e^{2\alpha h} U_i^\top W L_i D_i & Q + h^2 e^{2\alpha h} U_i^\top W \\ * & * & -\frac{\pi^2}{4} W + h^2 e^{2\alpha h} D_i^\top L_i^\top W L_i D_i & h^2 e^{2\alpha h} D_i^\top L_i^\top W \\ * & * & * & -\gamma I_n + h^2 e^{2\alpha h} W \end{bmatrix} \quad (16)$$

$i \in [\ell]$, satisfying (15). Due to the exponential Wirtinger inequality (Fridman, 2014) and the fact that $s \mapsto \delta_e(s) = e(s) - e(s_k)$ is absolutely continuous with square-integrable derivative, the functional $V_e(t)$ satisfies

$$\begin{aligned} V_e(t) &\geq e(t)^\top Q e(t) \geq \mathcal{C}_3 \|e(t)\|^2, \quad t \geq 0, \\ V_e(s_k) &= e(s_k)^\top Q e(s_k) \leq \mathcal{C}_4 \|e(s_k)\|^2, \quad k \in \mathbb{N}_0 \end{aligned} \quad (25)$$

for some explicitly computable $\mathcal{C}_i > 0$, $i = 3, 4$. While $V_e(t)$ may exhibit jumps on $\{s_k\}_{k=1}^\infty$, the exponential Wirtinger inequality further guarantees

$$\lim_{t \rightarrow s_k^-} V_e(t) \geq V_e(s_k). \quad (26)$$

Also, since $t_{k+1} - t_k \geq T > 0$, the interval $[s_k, s_{k+1})$ intersects $\{t_k\}_{k=0}^\infty$ at a finite number of points, where $\dot{e}$ exhibits a jump discontinuity, due to switching in $\sigma(t)$. Thus, $\dot{e}$ is piecewise continuous on $[s_k, s_{k+1})$ and the function

$$t \mapsto \int_{s_k}^t e^{-2\alpha(t-s)} \dot{e}(s)^\top W \dot{e}(s) ds$$

is absolutely continuous with the derivative existing outside of a discrete set $\mathfrak{T}_1$ that has a finite intersection with any bounded set in $[0, \infty)$. Therefore, $V(t)$ is indeed a candidate Lyapunov functional, which is continuously differentiable outside the discrete set $\mathfrak{T} \cup \mathfrak{T}_1$, whose intersection with every bounded set in $[0, \infty)$ is finite.

We first consider t in the interval $[t_k, t_k + T)$, where we denote $\sigma(t) = i$. Almost everywhere in $[t_k, t_k + T)$, it holds

$$\begin{aligned} \dot{V}_\varphi(t) &= 2\dot{\varphi}(t)^\top P_i(t)\varphi(t) + \varphi(t)^\top \dot{P}_i(t)\varphi(t) \\ &= 2[A_i\varphi(t) + L_i D_i(e(t) - \delta_e(t))]^\top P_i(t)\varphi(t) \\ &\quad + \varphi(t)^\top \dot{P}_i(t)\varphi(t) \\ &= \varphi(t)^\top [A_i^\top P_i(t) + P_i(t)A_i + \dot{P}_i(t)]\varphi(t) \\ &\quad + 2\varphi(t)^\top P_i(t)[L_i D_i e(t) - L_i D_i \delta_e(t)] \\ &= -\varphi(t)^\top C^\top C \varphi(t) - 2\zeta \varphi(t)^\top P_i(t)\varphi(t) \\ &\quad + 2\varphi(t)^\top P_i(t)[L_i D_i e(t) - L_i D_i \delta_e(t)]. \end{aligned}$$

Therefore,

$$\begin{aligned} \dot{V}_\varphi(t) &+ 2\alpha V_\varphi(t) + \varphi(t)^\top C^\top C \varphi(t) \\ &= -2(\zeta - \alpha)\varphi(t)^\top P_i(t)\varphi(t) \\ &\quad + 2\varphi(t)^\top P_i(t)[L_i D_i e(t) - L_i D_i \delta_e(t)]. \end{aligned} \quad (27)$$

Differentiating $V_e(t)$ a.e. in $[t_k, t_k + T)$ yields

$$\begin{aligned} \dot{V}_e(t) &+ 2\alpha V_e(t) \\ &= 2e(t)^\top Q \dot{e}(t) + h^2 e^{2\alpha h} \dot{e}(t)^\top W \dot{e}(t) \\ &\quad + 2\alpha e(t)^\top Q e(t) - \frac{\pi^2}{4} \delta_e(t)^\top W \delta_e(t) \\ &= 2e(t)^\top Q [U_i e(t) + L_i D_i \delta_e(t) + f_i(\varphi(t) + e(t))] \\ &\quad + 2\theta e(t)^\top Q b_i + 2\alpha e(t)^\top Q e(t) \\ &\quad + h^2 e^{2\alpha h} \dot{e}(t)^\top W \dot{e}(t) - \frac{\pi^2}{4} \delta_e(t)^\top W \delta_e(t). \end{aligned}$$

As a consequence,

$$\begin{aligned} \dot{V}_e(t) &+ 2\alpha V_e(t) \\ &= e(t)^\top \Phi_i(\alpha, Q) e(t) + 2\theta e(t)^\top Q b_i \\ &\quad + 2e(t)^\top Q [L_i D_i \delta_e(t) + f_i(\varphi(t) + e(t))] \\ &\quad + h^2 e^{2\alpha h} \dot{e}(t)^\top W \dot{e}(t) - \frac{\pi^2}{4} \delta_e(t)^\top W \delta_e(t). \end{aligned} \quad (28)$$

For simplicity of notation, let us further write

$$\chi_i(t) := f_i(x(t)) = f_i(\varphi(t) + e(t)). \quad (29)$$

Then, in view of Assumption 1,

$$\begin{aligned} N_i(t) &:= \kappa^2 \begin{bmatrix} \varphi(t) \\ e(t) \end{bmatrix}^\top \begin{bmatrix} I_n & I_n \\ I_n & I_n \end{bmatrix} \begin{bmatrix} \varphi(t) \\ e(t) \end{bmatrix} - \|\chi_i(t)\|^2 \\ &= \kappa^2 \|\varphi(t) + e(t)\|^2 - \|\chi_i(t)\|^2 \\ &= \kappa^2 \|x(t)\|^2 - \|\chi_i(t)\|^2 \geq 0, \quad t \geq 0. \end{aligned} \quad (30)$$

Finally, employing (11), we have

$$\begin{aligned} \dot{e}(t)^\top W \dot{e}(t) &= [U_i e(t) + L_i D_i \delta_e(t) + \theta b_i + \chi_i(t)]^\top W \\ &\quad \times [U_i e(t) + L_i D_i \delta_e(t) + \theta b_i + \chi_i(t)] \\ &= \begin{bmatrix} e(t) \\ \delta_e(t) \\ \chi_i(t) \\ b_i \end{bmatrix}^\top \begin{bmatrix} U_i^\top \\ D_i^\top L_i^\top \\ I_n \\ \theta I_n \end{bmatrix} W \begin{bmatrix} U_i^\top \\ D_i^\top L_i^\top \\ I_n \\ \theta I_n \end{bmatrix} \begin{bmatrix} e(t) \\ \delta_e(t) \\ \chi_i(t) \\ b_i \end{bmatrix}. \end{aligned} \quad (31)$$

Recall (2) and denote

$$\xi(t) = \text{col}\{\varphi(t), e(t), \delta_e(t), \chi_i(t), b_i\}. \quad (32)$$

Combining (27), (28), (30) and (31), and letting $\gamma, \Upsilon > 0$, a.e. in $[t_k, t_k + T)$ we have

$$\begin{aligned} \dot{V}(t) &+ 2\alpha V(t) + \varphi(t)^\top C^\top C \varphi(t) - \theta \Upsilon \bar{b}^2 \\ &\leq \dot{V}(t) + 2\alpha V(t) + \varphi(t)^\top C^\top C \varphi(t) - \theta \Upsilon \|b_i\|^2 + \gamma N_i(t) \\ &= \xi(t)^\top \tilde{\Psi}_i(t) \xi(t), \end{aligned}$$

where

$$\tilde{\Psi}_i(t) = \begin{bmatrix} \Psi_i(t) & \begin{bmatrix} 0 \\ \theta(Q + h^2 e^{2\alpha h} U_i^\top W) \\ \theta h^2 e^{2\alpha h} D_i^\top L_i^\top W \\ \theta h^2 e^{2\alpha h} W \end{bmatrix} \\ * & -\theta(\Upsilon I_n + h^2 e^{2\alpha h} W) \end{bmatrix}, \quad t \in [t_k, t_k + T) \quad (33)$$

with $\Psi_i(t)$ given in (16). By Schur complement and taking $\Upsilon \rightarrow \infty$, it is clear that $\tilde{\Psi}_i(t) \prec 0$ for all $t \in [0, T]$ guarantees the existence of a large enough $\Upsilon > 0$ such that $\tilde{\Psi}(t) \prec 0$ for all $t \in [t_k, t_k + T)$. Thus, $\Psi_i(t) \prec 0$ for all $t \in [0, T]$ guarantees, for some $\Upsilon > 0$, that

$$\dot{V}(t) + 2\alpha V(t) + \varphi(t)^\top C^\top C \varphi(t) - \theta \Upsilon \bar{b}^2 \leq 0 \quad \text{a.e. in } [t_k, t_k + T).$$

Consider now the interval $[t_k + T, t_{k+1})$, where $\sigma(t) = i$ by (14), as the switching condition has not been activated. Taking the derivative of $V(t)$ a.e. in $[t_k + T, t_{k+1})$, we have that $\dot{V}_e(t)$ remains as in (28). On the other hand, the derivative of $V_\varphi(t) = \varphi(t)^\top X_i \varphi(t)$, yields

$$\begin{aligned}
\dot{V}_\varphi(t) &= 2\varphi(t)^\top X_i \dot{\varphi}(t) \\
&= 2\varphi(t)^\top X_i (A_i \varphi(t) + L_i D_i e(t) - L_i D_i \delta_e(t)) \\
&= \varphi(t)^\top (X_i A_i + A_i^\top X_i) \varphi(t) + 2\varphi(t)^\top X_i L_i D_i e(t) \\
&\quad - 2\varphi(t)^\top X_i L_i D_i \delta_e(t).
\end{aligned} \tag{34}$$

Since no switching has occurred on $[t_k + T, t_{k+1})$, (14b) implies that, for all $j \in [\ell] \setminus \{i\}$,

$$\varphi(t)^\top (Y_{1,j} + Y_{2,j} - X_i) \varphi(t) \geq 0, \quad t \in [t_k + T, t_{k+1}),$$

whence, taking into account that $\Pi \in \mathcal{M}_\ell$,

$$\begin{aligned}
\dot{V}_\varphi(t) &\leq \varphi(t)^\top \left(X_i A_i + A_i^\top X_i \right. \\
&\quad \left. + \sum_{j \in [\ell] \setminus \{i\}} \Pi_{ij} (Y_{1,j} + Y_{2,j} - X_i) \right) \varphi(t) \\
&\quad + 2\varphi(t)^\top X_i L_i D_i e(t) - 2\varphi(t)^\top X_i L_i D_i \delta_e(t) \\
&\leq -\varphi(t)^\top C^\top C \varphi(t) - 2\zeta \varphi(t)^\top X_i \varphi(t) \\
&\quad + 2\varphi(t)^\top X_i L_i D_i e(t) - 2\varphi(t)^\top X_i L_i D_i \delta_e(t).
\end{aligned}$$

As a consequence,

$$\begin{aligned}
\dot{V}_\varphi(t) + 2\alpha V_\varphi(t) + \varphi(t)^\top C^\top C \varphi(t) \\
\leq -2(\zeta - \alpha) \varphi(t)^\top X_i \varphi(t) + 2\varphi(t)^\top X_i L_i D_i e(t) \\
- 2\varphi(t)^\top X_i L_i D_i \delta_e(t) \quad \text{a.e. in } [t_k + T, t_{k+1}).
\end{aligned} \tag{35}$$

Recalling (15b), the upper bound in (35) can be obtained by replacing $P_i(t)$ in the right-hand side of (27) with $P_i(T) = X_i$. Taking into account (35) and applying arguments similar to (28)-(32), we have

$$\begin{aligned}
\dot{V}(t) + 2\alpha V(t) \\
+ \varphi(t)^\top C^\top C \varphi(t) - \theta \Upsilon \bar{b}^2 \leq \xi(t)^\top \tilde{\Psi}_i(T) \xi(t)
\end{aligned} \tag{36}$$

a.e. in $[t_k + T, t_{k+1})$, where we employ $P_i(T) = X_i$. Thus, $\Psi_i(t) < 0$, $t \in [0, T]$, with $\Psi_i(t)$ given in (16), guarantees

$$\begin{aligned}
\dot{V}(t) + 2\alpha V(t) + \varphi(t)^\top C^\top C \varphi(t) - \theta \Upsilon \bar{b}^2 \leq 0, \\
\text{a.e. in } [t_k + T, t_{k+1}).
\end{aligned}$$

We now show that $V(t)$ does not exhibit upward jumps at the instants $\{t_k\}_{k=1}^\infty$. By (21), since $V_e(t)$ exhibits only downward jumps, whenever it is discontinuous, as per (26), it suffices to show that $V_\varphi(t)$ does not exhibit upward jumps on $\{t_k\}_{k=1}^\infty$ (recall that $t_0 = 0$, by assumption). Let $k \in \mathbb{N}_0$ and assume that $\sigma(t_{k+1}^-) = i$ and $\sigma(t_{k+1}) = j$. Due to continuity of $\varphi(t)$, it holds that

$$\begin{aligned}
V_\varphi(t_{k+1}^-) &= \varphi(t_{k+1})^\top X_i \varphi(t_{k+1}), \\
V_\varphi(t_{k+1}) &= \varphi(t_{k+1})^\top P_j(t_{k+1}) \varphi(t_{k+1}).
\end{aligned}$$

By integrating (15), we have $P_j(t_{k+1}) = Y_{1,j} + Y_{2,j}$. Hence, $V_\varphi(t_{k+1}) - V_\varphi(t_{k+1}^-) = \varphi(t_{k+1})^\top [Y_{1,j} + Y_{2,j} - X_i] \varphi(t_{k+1})$. Recalling the switching condition (14c), at $t = t_{k+1}$ there exists $j_* \in [\ell] \setminus \{i\}$ such that

$$\varphi(t_{k+1})^\top [Y_{1,j_*} + Y_{2,j_*} - X_i] \varphi(t_{k+1}) \leq 0$$

and

$$\sigma(t_{k+1}) = \underset{j \in [\ell] \setminus \{i\}}{\operatorname{argmin}} \varphi(t_{k+1})^\top [Y_{1,j} + Y_{2,j}] \varphi(t_{k+1}).$$

Therefore, recalling that $\sigma(t_{k+1}) = j$,

$$\begin{aligned}
\varphi(t_{k+1})^\top [Y_{1,j} + Y_{2,j}] \varphi(t_{k+1}) \\
\leq \varphi(t_{k+1})^\top [Y_{1,j_*} + Y_{2,j_*}] \varphi(t_{k+1}) \leq \varphi(t_{k+1})^\top X_i \varphi(t_{k+1}),
\end{aligned}$$

implying

$$V_\varphi(t_{k+1}) - V_\varphi(t_{k+1}^-) \leq 0, \tag{37}$$

whence also V_φ cannot exhibit upward jumps.

Summarising, we have that

$$\begin{aligned}
\dot{V}(t) + 2\alpha V(t) + \varphi(t)^\top C^\top C \varphi(t) - \theta \Upsilon \bar{b}^2 \leq 0, \\
\text{a.e. in } [t_k, t_{k+1}),
\end{aligned} \tag{38}$$

and functional $V(t)$ exhibits no upward jumps at $\{t_k\}_{k=1}^\infty$. Thus, for the case $\theta = 0$, multiplying (38) by $e^{2\alpha t}$ and integrating it from t_k to t , with $t \in [t_k, t_{k+1})$, yields

$$e^{2\alpha t} V(t) - e^{2\alpha t_k} V(t_k) + \int_{t_k}^t e^{2\alpha s} \varphi(s)^\top C^\top C \varphi(s) ds \leq 0.$$

Summing the latter inequality over $[t_{k'}, t_{k'+1}]$, with $k' = 0, \dots, k-1$ and $t_0 = 0$, and $[t_k, t]$ yields

$$e^{2\alpha t} V(t) + \int_0^t e^{2\alpha s} \varphi(s)^\top C^\top C \varphi(s) ds \leq V(0). \tag{39}$$

In particular, we obtain $V(t) \leq e^{-2\alpha t} V(0)$ for all $t \geq 0$. Recalling (24) and (25), and employing $\varphi(0) = 0$ and $e(0) = x(0)$, the exponential decay of $V(t)$ yields (17) with $M := \frac{C_4}{\min(\mathcal{C}_1, \mathcal{C}_3)}$, which is explicitly computable. Control Objective 1 is thus fulfilled for $\theta = 0$. Furthermore, consider the average cost estimate $\hat{J}(t)$ defined in (18), which is bounded on $[0, \infty)$. Indeed, (39) implies that $\int_0^t \varphi(s)^\top C^\top C \varphi(s) ds \leq V(0)$; then, recalling that $V(0) = x(0)^\top Q x(0)$ and taking $t \rightarrow \infty$, we have that $\sup_{t \geq 0} t \hat{J}(t) \leq x(0)^\top Q x(0)$. Assume now that $\|x(0)\| \leq R$ for some $R > 0$. Then, letting $\tau > 0$, we have

$$\begin{aligned}
J(\tau) &= \frac{1}{\tau} \int_0^\tau x(s)^\top C^\top C x(s) ds \\
&= \frac{1}{\tau} \int_0^\tau (\varphi(s)^\top + e(s)^\top) C^\top C (\varphi(s) + e(s)) ds \\
&= \hat{J}(\tau) + \frac{2}{\tau} \int_0^\tau e(s)^\top C^\top C \varphi(s) ds \\
&\quad + \frac{1}{\tau} \int_0^\tau e(s)^\top C^\top C e(s) ds.
\end{aligned} \tag{40}$$

Taking into account (17), we have

$$\begin{aligned}
|J(\tau) - \hat{J}(\tau)| &\leq \frac{3M \|C^\top C\|}{2\alpha\tau} \|x(0)\|^2 (1 - e^{-2\alpha\tau}) \\
&\leq \frac{3M \|C^\top C\| R^2}{2\alpha\tau},
\end{aligned}$$

where the constants in the final upper bound are computable explicitly. Hence, given $\varepsilon > 0$, one can compute a time $t_* = t_*(\varepsilon, R)$ such that $t \geq t_*$ implies $|J(t) - \hat{J}(t)| < \varepsilon$. Hence, $\hat{J}(t)$ fulfils control Objective 2.

Similarly, for the case $\theta = 1$, (38) yields

$$\dot{V}(t) + 2\alpha V(t) + \varphi(t)^\top C^\top C \varphi(t) - \Upsilon \bar{b}^2 \leq 0, \tag{41}$$

whence arguments similar to those leading to (39) yield

$$V(t) \leq e^{-2\alpha t} V(0) + \frac{\Upsilon \bar{b}^2}{2\alpha}. \tag{42}$$

Using (24) and (25), we obtain, for $t \geq 0$,

$$\|\varphi(t)\|^2 + \|e(t)\|^2 \leq M e^{-2\alpha t} \|x(0)\|^2 + \frac{\Upsilon \bar{b}^2}{2\alpha \min(\mathcal{C}_1, \mathcal{C}_3)},$$

with $M := \frac{C_4}{\min(\mathcal{C}_1, \mathcal{C}_3)}$. Since $\|x(t)\|^2 \leq 2\|\varphi(t)\|^2 + 2\|e(t)\|^2$, the explicit ultimate bounds (19) follow. Control Objective 1 is thus fulfilled for $\theta = 1$. Furthermore, since both $\|e(t)\|$ and $\|\varphi(t)\|$ are bounded via $\bar{B} + o(1)$, $t \rightarrow \infty$, with $\bar{B}$ given in (19), employing (40) finally yields (20). ■

Remark 7. If one is interested in the stabilisation or ultimate-boundedness of affine switched systems, subject to dwell time, the inequality (16) with $\kappa = 0$ provides sufficient

conditions to achieve this control objective. Indeed, since $f_i(0) = 0$ for all $i \in [\ell]$, in view of Assumption 1, setting the Lipschitz constant $\kappa = 0$ is equivalent to setting $f_i \equiv 0$ for all $i \in [\ell]$. Similarly, setting $h = 0$ in (16) can be shown to recover the case of continuous-time measurements of the output $y(t)$, $t \geq 0$.

Remark 8. Theorem 1 can also be applied to the design of an observer-based switching strategy, subject to minimum dwell-time, for the system

$$\begin{cases} \dot{x}(t) = A_{\sigma(t)}x(t) + g_{\sigma(t)}(x(t)), & x(t_0) = x_0, \\ y(t) = D_{\sigma(t)}x(s_k), & t \in [s_k, s_{k+1}), \end{cases} \quad (43)$$

where $g_{\sigma(t)} \in \{g_j\}_{j \in [\ell]}$ and g_j , $j \in [\ell]$, are globally Lipschitz. Indeed, system (43) can be presented as (1) with $\theta = 1$, $b_{\sigma(t)} = g_{\sigma(t)}(0)$ and $f_{\sigma(t)}(x) = g_{\sigma(t)}(x) - g_{\sigma(t)}(0)$, with the latter being globally Lipschitz and equal to zero at the origin. If $g_j(0)$, $j \in [\ell]$, are unknown, but $\max_{i \in [\ell]} \|g_i(0)\|$ can be upper bounded, Theorem 1 can be applied to obtain ultimate boundedness of the closed-loop system (43), with an explicit estimate of the ultimate bound. If $g_j(0)$, $j \in [\ell]$, are known, we refer the reader to the modified switching law design of Section 3.4.

3.3 Design of the observer gains $\{L_i\}_{i \in [\ell]}$

Theorem 1 assumes that the gains L_i , $i \in [\ell]$, are given and proceeds to obtain $Q \succ 0$ such that $\Phi_i(\alpha, Q) \prec 0 \forall i \in [\ell]$; see the (2, 2) block in (16). The design of the observer gains $\{L_i\}_{i \in [\ell]}$ via the approach in Remark 2 may lead to significant conservatism in verifying $\Psi_i(t) \prec 0$, $i \in [\ell]$, $t \in [0, T]$, with $\Psi_i(t)$ given in (16). Indeed, such a design of $\{L_i\}_{i \in [\ell]}$ does not take into account that the obtained gains are to be substituted into $\Psi_i(t) \prec 0$, $i \in [\ell]$, $t \in [0, T]$ as exogenous parameters and thereby does not factor the presence of the other decision variables Q and W , as well as the matrices X_i , coming from (12), in the inequalities to be verified.

Thus, it is of interest to design the observer gains from the matrix inequalities obtained in Theorem 1. The following proposition, where we assume for simplicity of presentation that $f_i \equiv 0$ for all $i \in [\ell]$, provides *sufficient* conditions for the design of L_i , $i \in [\ell]$ via Bilinear Matrix Inequalities (BMIs). BMIs, known to be $\mathcal{NP}$ -hard (Toker and Ozbay, 1995), can be solved using, e.g., Alternating Convex Search and Augmented Lagrangian Methods (see Boyd and Vandenberghe, 2004 and VanAntwerp and Braatz, 2000), which are implemented in existing software packages, including PENBMI (see Kočvara and Stingl, 2003 and references therein) and BMISOLVER (see Tran Dinh et al., 2012a,b and references therein).

Proposition 1. Let $\zeta, T > 0$, $\{X_i\}_{i \in [\ell]} \subseteq \mathbb{R}^{n \times n}$, $X_i \succ 0$, $i \in [\ell]$, and $\Pi \in \mathcal{M}_\ell$ satisfy the Lyapunov-Metzler inequality (12), and consider the time-varying symmetric positive-definite matrices $P_i: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{n \times n}$ satisfying (15). Given tuning parameters $h > 0$, $\alpha > 0$, let there exist matrices $Q \succ 0$, $W \succ 0$, $R \succ 0$, $Y_i \in \mathbb{R}^{n \times p}$, $i \in [\ell]$, and $J \in \mathbb{R}^{n \times 2n}$ such that

$$\begin{bmatrix} R & 0 \\ 0 & -W \end{bmatrix} + \begin{bmatrix} I_n \\ -W \end{bmatrix} J + J^\top \begin{bmatrix} I_n & -W \end{bmatrix} \prec 0, \quad (44)$$

$$\Lambda_i(t) \prec 0, \quad i \in [\ell], \quad t \in [0, T],$$

where $\{\Lambda_i(t)\}_{i \in [\ell]}$ are given in (47). Then, setting the observer gains as $L_i = Q^{-1}Y_i$, the inequalities $\Psi_i(t) \prec 0$, $i \in [\ell]$, $t \in [0, T]$, with $\{\Psi_i(t)\}_{i \in [\ell]}$ given in (16), hold with L_i , $i \in [\ell]$ and the same values of ζ , α , h , W , Q .

Remark 9. The inequalities $\Lambda_i(t) \prec 0$, $i \in [\ell]$, $t \in [0, T]$ are linear in Q , Y_i , $i \in [\ell]$, W and R . Thus, the only source of non-linearity in (44) arises from the first inequality, which is quadratic in W and J . Furthermore, as the proof of Proposition 1 shows, only the matrix $W \succ 0$ prevents the conversion of $\Psi_i(t) \prec 0$, $i \in [\ell]$, $t \in [0, T]$ into LMIs for the design of the gains. Recall that W emanates from the component of $V_e(t)$ in (23), which is aimed at compensating the sampling in the state measurements.

Proof. Existence of $J \in \mathbb{R}^{n \times 2n}$ such that the first inequality in (44) holds implies $W^{-1} \succ R$, i.e., $\vartheta^\top W^{-1} \vartheta > \vartheta^\top R \vartheta$ for any $\vartheta \in \mathbb{R}^n \setminus \{0\}$. In fact, $W^{-1} \succ R$ is equivalent to

$$\zeta^\top H^\top H \zeta = 0, \quad \zeta \neq 0 \implies \zeta^\top G \zeta < 0, \quad (45)$$

where

$$H = \begin{bmatrix} I & -W \end{bmatrix}, \quad G = \begin{bmatrix} R & 0 \\ 0 & -W \end{bmatrix}, \quad \zeta = \begin{bmatrix} \vartheta \\ z \end{bmatrix}.$$

By Finsler's lemma (Schweppe, 1973), (45) is equivalent to the existence of $\mu \in \mathbb{R}$ such that $G - \mu H^\top H \prec 0$. Employing (Boyd et al., 1994, Section 2.6.2), the latter holds iff there exists $J \in \mathbb{R}^{n \times 2n}$ such that

$$G + H^\top J + J^\top H \prec 0,$$

which is exactly the first inequality in (44).

Recall (31) in the proof of Theorem 1, and denote $Y_i = QL_i$. In view of the assumption that $f_i \equiv 0$, $i \in [\ell]$, we can consider $\Psi_i(t)$ in (16) with the last row and the last column deleted, as well as set $\kappa = 0$. Performing Schur complement, the inequality $\Psi_i(t) \prec 0$, $t \in [0, T]$ is then equivalent to $\Psi_i^{(1)}(t; W^{-1}) \prec 0$, $t \in [0, T]$, with $\Psi_i^{(1)}(t; W^{-1})$ given in (47). In view of $W^{-1} \succ R$, which has been previously demonstrated, we have $-h^{-2}e^{-2\alpha h}W^{-1} \prec -h^{-2}e^{-2\alpha h}R$, whence $\Psi_i^{(1)}(t; W^{-1}) \preceq \Psi_i^{(1)}(t; R)$, $t \in [0, T]$, where $\Psi_i^{(1)}(t; R)$ is obtained from $\Psi_i^{(1)}(t; W^{-1})$ by replacing W^{-1} with R in the (4, 4) entry.

Next, denoting

$$Z_1 = \begin{bmatrix} P_i(t) \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad Z_2 = \begin{bmatrix} 0 \\ D_i^\top Y_i^\top \\ -D_i^\top Y_i^\top \\ 0 \end{bmatrix}, \quad Z_3 = -Z_2, \quad Z_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ I \end{bmatrix}, \quad (46)$$

we see that

$$\begin{aligned} \Psi_i^{(1)}(t; R) &= \Psi_i^{(2)}(t; R) \\ &\quad + \sum_{i=1}^2 (Z_{2i-1}Q^{-1}Z_{2i}^\top + Z_{2i}Q^{-1}Z_{2i-1}^\top), \end{aligned}$$

with $\Psi_i^{(2)}(t; R)$ given in (47). Performing a square completion, for $i = 1, 2$ we have

$$\begin{aligned} &Z_{2i-1}Q^{-1}Z_{2i}^\top + Z_{2i}Q^{-1}Z_{2i-1}^\top \\ &= \frac{1}{2} (Z_{2i-1} + Z_{2i}) Q^{-1} (Z_{2i-1} + Z_{2i})^\top \\ &\quad - \frac{1}{2} (Z_{2i-1} - Z_{2i}) Q^{-1} (Z_{2i-1} - Z_{2i})^\top, \end{aligned}$$

with the rightmost term being negative semidefinite. Thus, we obtain

$$\Psi_i^{(1)}(t; R) \leq \Psi_i^{(2)}(t; R) + \frac{1}{2} \sum_{i=1}^2 (Z_{2i-1} + Z_{2i}) Q^{-1} (Z_{2i-1} + Z_{2i})^\top.$$

By applying Schur complement to the right-hand side, we conclude that $\Psi_i^{(1)}(t; R) \prec 0$, $i \in [\ell]$, $t \in [0, T]$ holds if $\Lambda_i(t) \prec 0$, $i \in [\ell]$, $t \in [0, T]$, with $\Lambda_i(t)$ given in (47). ■

The BMI-based design procedure given in Proposition 1 may be computationally challenging. However, the proof of Proposition 1 suggests two ways in which such a design may be relaxed in order to recover LMIs for the design of the observer gains $\{L_i\}_{i \in [\ell]}$.

Proposition 2. Let $\zeta, T > 0$, $\{X_i\}_{i \in [\ell]} \subseteq \mathbb{R}^{n \times n}$, $X_i \succ 0$, $i \in [\ell]$, and $\Pi \in \mathcal{M}_\ell$ satisfy the Lyapunov-Metzler inequality (12), and consider the time-varying symmetric positive-definite matrices $P_i: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{n \times n}$ satisfying (15). Given tuning parameters $h > 0$, $\alpha > 0$, let there exist matrices $Q \succ 0$ and $Y_i \in \mathbb{R}^{n \times p}$ such that $\Theta_i(t) \prec 0$, $i \in [\ell]$, $t \in [0, T]$, with $\{\Theta_i(t)\}_{i \in [\ell]}$ given in (48). Then, setting the observer gains as $L_i = Q^{-1}Y_i$, the inequalities $\Psi_i(t) \prec 0$, $i \in [\ell]$, $t \in [0, T]$, with $\{\Psi_i(t)\}_{i \in [\ell]}$ given in (16), hold with L_i , $i \in [\ell]$ and the same values of ζ , α , h , Q and $W = Q$.

Proof. The proof follows arguments similar to those of Proposition 1. Thus, we only provide a sketch.

In view of the assumption that $f_i \equiv 0$, $i \in [\ell]$, we can consider $\Psi_i(t)$ in (16) with the last row and the last column deleted, as well as set $\kappa = 0$. Then, setting $W = Q$ and $Y = QL_i$, we note that $\Psi_i(t)$ becomes $\tilde{\Psi}_i(t)$, given in (48). We further have

$$\begin{aligned} U_i^\top Q U_i &= (Q A_i - Y_i D_i)^\top Q^{-1} (Q A_i - Y_i D_i), \\ U_i^\top Q L_i D_i &= (Q A_i - Y_i D_i)^\top Q^{-1} Y_i D_i, \\ D_i^\top L_i^\top Q L_i D_i &= D_i^\top Y_i^\top Q^{-1} Y_i D_i. \end{aligned}$$

Thus,

$$\tilde{\Psi}_i(t) = \Theta_i^0(t) + Z_1 Q^{-1} Z_2^\top + Z_2 Q^{-1} Z_1^\top + h^2 e^{2\alpha h} V_1 Q^{-1} V_1^\top,$$

where Z_1 and Z_2 are as in (46), but with the last entry deleted, and

$$V_1 = \begin{bmatrix} 0 \\ (Q A_i - Y_i D_i)^\top \\ D_i^\top Y_i^\top \end{bmatrix}.$$

The result of the proposition now follows by applying a square completion, followed by Schur complement, similarly to the proof of Proposition 1. ■

An alternative design approach, which leads to even simpler LMIs, is as follows. Considering $\Psi_i(t)$ in (16) and set $h = 0$, $\alpha = 0$, $\kappa = 0$ and $Y_i = QL_i$, $i \in [\ell]$, it can be easily seen that $\Psi_i(t) \prec 0$, $i \in [\ell]$ hold iff

$$\begin{bmatrix} -2\zeta P_i(t) & P_i(t) Q^{-1} Y_i D_i \\ \star & A_i^\top Q + Q A_i - D_i^\top Y_i^\top - Y_i D_i \end{bmatrix} \prec 0, \quad (49)$$

$$i \in [\ell], \quad t \in [0, T].$$

By performing square completion as in the proof of Proposition 1, feasibility of (49) is guaranteed if

$$\begin{bmatrix} -2\zeta P_i(t) & 0 & P_i(t) \\ \star & A_i^\top Q + Q A_i - D_i^\top Y_i^\top - Y_i D_i & D_i^\top Y_i^\top \\ \star & \star & -2Q \end{bmatrix} \prec 0, \quad (50)$$

$$i \in [\ell], \quad t \in [0, T].$$

The LMIs (50) can be used to obtain the gains $L_i = Q^{-1}Y_i$, $i \in [\ell]$. Then, the obtained gains can be employed in verifying $\Psi_i(t) \prec 0$, $i \in [\ell]$, $t \in [0, T]$ with $h > 0$, $\alpha > 0$ and $\kappa > 0$.

3.4 Modified switching law for the case of known b_i , $i \in [\ell]$

The analysis of Section 3.1 assumes that the affine terms $\{b_i\}_{i \in [\ell]}$ are unknown and only the bound (2) is available. As a result, the observer design in (9) and the proposed switching law (14) do not take into account the presence of $\{b_i\}_{i \in [\ell]}$ in the dynamics of (1) when $\theta = 1$. If $\{b_i\}_{i \in [\ell]}$ are known, simply employing Theorem 1 in order to obtain an estimate on the ultimate bound may lead to conservative results, due to inefficient tracking of $x(t)$ by $\varphi(t)$ and unnecessary switches of the switching law (14). In this section, we propose a modified switching law tailored to the case where $\{b_i\}_{i \in [\ell]}$ are known.

Consider (1) with $\theta = 1$ and $\{b_i\}_{i \in [\ell]}$ known and let

$$\begin{aligned} \bar{x}(t) &= \begin{bmatrix} x(t) \\ 1 \end{bmatrix} \in \mathbb{R}^{n+1}, \quad \bar{C} = [C \mid 0] \in \mathbb{R}^{p \times (n+1)}, \\ \bar{A}_{\sigma(t)} &= \begin{bmatrix} A_{\sigma(t)} & b_{\sigma(t)} \\ 0 & 0 \end{bmatrix} \in \mathbb{R}^{(n+1) \times (n+1)}, \\ \bar{f}_{\sigma(t)}(\bar{x}(t)) &= \begin{bmatrix} f_{\sigma(t)}(x(t)) \\ 0 \end{bmatrix} \in \mathbb{R}^{n+1}, \\ \bar{D}_{\sigma(t)} &= [D_{\sigma(t)} \mid 0] \in \mathbb{R}^{m \times (n+1)}. \end{aligned}$$

Then, (1) can be embedded into the extended system

$$\begin{aligned} \dot{\bar{x}}(t) &= \bar{A}_{\sigma(t)} \bar{x}(t) + \bar{f}_{\sigma(t)}(\bar{x}(t)), \quad \bar{x}(t_0) = \begin{bmatrix} x_0 \\ 1 \end{bmatrix}, \\ y(t) &= \bar{D}_{\sigma(t)} \bar{x}(s_k), \quad t \in [s_k, s_{k+1}), \\ z(t) &= \bar{C} \bar{x}(t), \end{aligned} \quad (51)$$

where $\{\bar{A}_i\}_{i \in [\ell]}$ are known, due to our assumption on availability of $\{b_i\}_{i \in [\ell]}$. In view of (51), let

$$\bar{\varphi}(t) = \begin{bmatrix} \varphi(t) \\ 1 \end{bmatrix} \in \mathbb{R}^{n+1}, \quad \bar{L}_{\sigma(t)} = \begin{bmatrix} L_{\sigma(t)} \\ 0 \end{bmatrix} \in \mathbb{R}^{(n+1) \times m},$$

where the observer $\bar{\varphi}(t)$ satisfies

$$\begin{aligned} \dot{\bar{\varphi}}(t) &= \bar{A}_{\sigma(t)} \bar{\varphi}(t) + \bar{L}_{\sigma(t)} [y(s_k) - \bar{D}_{\sigma(t)} \bar{\varphi}(s_k)], \\ t &\in [s_k, s_{k+1}), \\ \bar{\varphi}(0) &= \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \end{aligned} \quad (52)$$

Here, $\{b_i\}_{i \in [\ell]}$ are included in the observer design through $\{\bar{A}_i\}_{i \in [\ell]}$, whereas the observer gains $\{L_i\}_{i \in [\ell]}$ satisfy Assumption 2. We denote

$$\begin{aligned} \bar{e}(t) &= \bar{x}(t) - \bar{\varphi}(t) = \begin{bmatrix} e(t) \\ 0 \end{bmatrix} = \begin{bmatrix} x(t) - \varphi(t) \\ 0 \end{bmatrix}, \\ \bar{\delta}_e(t) &= \begin{bmatrix} \delta_e(t) \\ 0 \end{bmatrix} = \bar{e}(t) - \bar{e}(s_k), \\ \bar{U}_{\sigma(t)} &= \bar{A}_{\sigma(t)} - \bar{L}_{\sigma(t)} \bar{D}_{\sigma(t)}. \end{aligned} \quad (53)$$

Then, by arguments similar to (10)–(11), we obtain the following closed-loop system for $(\bar{\varphi}(t), \bar{e}(t))$

$$\begin{aligned} \dot{\bar{\varphi}}(t) &= \bar{A}_{\sigma(t)} \bar{\varphi}(t) + \bar{L}_{\sigma(t)} \bar{D}_{\sigma(t)} \bar{e}(t) - \bar{L}_{\sigma(t)} \bar{D}_{\sigma(t)} \bar{\delta}_e(t), \\ \dot{\bar{e}}(t) &= \bar{U}_{\sigma(t)} \bar{e}(t) + \bar{L}_{\sigma(t)} \bar{D}_{\sigma(t)} \bar{\delta}_e(t) + \bar{f}_{\sigma(t)}(\bar{\varphi}(t) + \bar{e}(t)). \end{aligned} \quad (54)$$

$$\Psi_i^{(1)}(t; W^{-1}) := \begin{bmatrix} -2(\zeta - \alpha)P_i(t) & P_i(t)Q^{-1}Y_iD_i & -P_i(t)Q^{-1}Y_iD_i & 0 \\ * & A_i^\top Q + QA_i - D_i^\top Y_i^\top - Y_iD_i + 2\alpha Q & Y_iD_i & A_i^\top - D_i^\top Y_i^\top Q^{-1} \\ * & * & -\frac{\pi^2}{4}W & D_i^\top Y_i^\top Q^{-1} \\ * & * & * & -h^{-2}e^{-2\alpha h}W^{-1} \end{bmatrix},$$

$$\Psi_i^{(2)}(t; R) := \begin{bmatrix} -2(\zeta - \alpha)P_i(t) & 0 & 0 & 0 \\ * & A_i^\top Q + QA_i - D_i^\top Y_i^\top - Y_iD_i + 2\alpha Q & Y_iD_i & A_i^\top \\ * & * & -\frac{\pi^2}{4}W & 0 \\ * & * & * & -h^{-2}e^{-2\alpha h}R \end{bmatrix}, \quad \Lambda_i(t) = \left[\begin{array}{c|c} \Psi_i^{(2)}(t; R) & \begin{bmatrix} P_i(t) & 0 \\ D_i^\top Y_i^\top & -D_i^\top Y_i^\top \\ -D_i^\top Y_i^\top & D_i^\top Y_i^\top \\ 0 & I \end{bmatrix} \\ \hline * & -2 \text{diag}(Q, Q) \end{array} \right]. \quad (47)$$

$$\tilde{\Psi}_i(t) := \begin{bmatrix} -2(\zeta - \alpha)P_i(t) & P_i(t)L_iD_i & -P_i(t)L_iD_i \\ * & \Phi_i(\alpha, Q) + h^2e^{2\alpha h}U_i^\top QU_i & QL_iD_i + h^2e^{2\alpha h}U_i^\top QL_iD_i \\ * & * & -\frac{\pi^2}{4}Q + h^2e^{2\alpha h}D_i^\top L_i^\top QL_iD_i \end{bmatrix}, \quad \Theta_i^1(t) = \begin{bmatrix} P_i(t) & 0 \\ D_i^\top Y_i^\top & A_i^\top Q - D_i^\top Y_i^\top \\ -D_i^\top Y_i^\top & D_i^\top Y_i^\top \end{bmatrix}, \quad (48)$$

$$\Theta_i^0(t) := \begin{bmatrix} -2(\zeta - \alpha)P_i(t) & 0 & 0 \\ * & A_i^\top Q + QA_i - D_i^\top Y_i^\top - Y_iD_i + 2\alpha Q & Y_iD_i \\ * & * & -\frac{\pi^2}{4}Q \end{bmatrix}, \quad \Theta_i(t) = \left[\begin{array}{c|c} \Theta_i^0(t) & \Theta_i^1(t) \\ \hline * & \begin{bmatrix} -2Q & 0 \\ * & -2h^{-2}e^{-2\alpha h}Q \end{bmatrix} \end{array} \right].$$

For the switching strategy design, consider given $\varepsilon, \zeta > 0$, matrices $\{X_i\}_{i \in [\ell]} \subseteq \mathbb{R}^{n \times n}$, $X_i \succ 0$, $i \in [\ell]$ and an irreducible Metzler matrix $\Pi \in \mathcal{M}_\ell$. We further denote

$$\bar{X}_i = \begin{bmatrix} X_i & 0 \\ 0 & 1 \end{bmatrix}, \quad E = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}. \quad (55)$$

Let the following Lyapunov-Metzler inequalities be satisfied

$$\bar{A}_i^\top \bar{X}_i + \bar{X}_i \bar{A}_i + \sum_{j \in [\ell] \setminus \{i\}} \Pi_{ij} (\bar{Y}_{1,j} + \bar{Y}_{2,j} - \bar{X}_i) + \bar{C}^\top \bar{C} + 2\zeta \bar{X}_i \prec \varepsilon E, \quad (56)$$

where $\bar{Y}_{1,j}$ and $\bar{Y}_{2,j}$ are given by the expressions in (13), with A_j , X_j and C replaced by $\bar{A}_j$, $\bar{X}_j$ and $\bar{C}$, respectively. Then, the observer-based switching law, with dwell time $T > 0$, is defined as follows. Given $t_k \in [0, \infty)$,

$$\sigma(t) = i, \quad t \in [t_k, t_k + T], \quad (57a)$$

$$\sigma(t) = i, \quad t > t_k + T, \quad \text{as long as, } \forall j \in [\ell] \setminus \{i\}, \quad (57b)$$

$$\text{it is } \bar{\varphi}(t)^\top (\bar{Y}_{1,j} + \bar{Y}_{2,j}) \bar{\varphi}(t) \geq \bar{\varphi}(t)^\top \bar{X}_i \bar{\varphi}(t),$$

$$\sigma(t_{k+1}) = \underset{j \in [\ell] \setminus \{i\}}{\text{argmin}} [\bar{\varphi}(t_{k+1})^\top (\bar{Y}_{1,j} + \bar{Y}_{2,j}) \bar{\varphi}(t_{k+1})], \quad (57c)$$

where

$$t_{k+1} := \inf_{t > t_k + T} \{t \mid \exists j: \bar{\varphi}(t)^\top [\bar{Y}_{1,j} + \bar{Y}_{2,j} - \bar{X}_i] \bar{\varphi}(t) < 0\}.$$

We now state the main theorem of this section, which is analogous to Theorem 1 for the case of known $\{b_i\}_{i \in [\ell]}$.

Theorem 2. Consider the closed-loop system (54) under Assumptions 1 and 2, with gains L_i fixed (e.g., as in Remark 2). Let $\varepsilon, \zeta, T > 0$, $\{X_i\}_{i \in [\ell]} \subseteq \mathbb{R}^{n \times n}$, $X_i \succ 0$, $i \in [\ell]$, and $\Pi \in \mathcal{M}_\ell$ satisfy the Lyapunov-Metzler inequality (55), (56) and consider the time-varying symmetric positive-definite matrices $\bar{P}_i: \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^{n \times n}$ such that

$$-\dot{\bar{P}}_i(t) = \bar{A}_i^\top \bar{P}_i(t) + \bar{P}_i(t) \bar{A}_i + \bar{C}^\top \bar{C} + 2\zeta \bar{P}_i(t), \quad t \in [t_k, t_k + T], \quad i \in [\ell], \quad (58a)$$

$$\bar{P}_i(t) = \bar{X}_i, \quad t \in [t_k + T, t_{k+1}), \quad i \in [\ell]. \quad (58b)$$

Given tuning parameters $h > 0$, $\alpha > 0$ and $\kappa > 0$, let there exist matrices $Q \succ 0$, $W \succ 0$ and scalar $\gamma > 0$ such that the LMIs $\tilde{\Psi}_i(t) \prec 0$ hold for all $i \in [\ell]$ and all $t \in [0, T]$,

where $\{\tilde{\Psi}_i(t)\}_{i \in [\ell]}$ are given in (59). Then, the observer-based switching law (57) guarantees that the solutions of the closed-loop system (54) satisfy the ultimate bounds

$$\limsup_{t \rightarrow \infty} (\|\varphi(t)\|^2 + \|e(t)\|^2) \leq \frac{\varepsilon}{2\alpha \min(\mathcal{C}_1, \mathcal{C}_3)} =: \bar{B}_1, \quad (60)$$

$$\limsup_{t \rightarrow \infty} \|x(t)\|^2 \leq 2\bar{B}_1,$$

with explicitly computable positive constants $\mathcal{C}_1$ and $\mathcal{C}_3$. Furthermore, the estimate $\hat{J}(t)$ of the average cost $J(t)$ satisfies (20) with $\bar{B}$ replaced by $\bar{B}_1$.

Proof. The proof follows the same arguments as the proof of Theorem 1. Thus, we only elaborate on the main differences.

Let $Q, W \in \mathbb{R}^{n \times n}$ with $Q, W \succ 0$ and $\alpha \in (0, \zeta)$ and consider the candidate Lyapunov functional

$$V(t) = V_{\bar{\varphi}}(t) + V_e(t), \quad (61)$$

where, in view of the structure of $\bar{e}(t)$ in (53), $V_e(t)$ is given by (23) and $V_{\bar{\varphi}}(t)$ is given by

$$V_{\bar{\varphi}}(t) = \bar{\varphi}(t)^\top \bar{P}_{\sigma(t)}(t) \bar{\varphi}(t), \quad t \geq 0. \quad (62)$$

Consider the interval $[t_k, t_k + T)$, where we denote $\sigma(t) = i$. By arguments similar to (27)–(28), almost everywhere on $[t_k, t_k + T)$, it holds

$$\begin{aligned} \dot{V}_{\bar{\varphi}}(t) + 2\alpha V_{\bar{\varphi}}(t) + \bar{\varphi}(t)^\top \bar{C}^\top \bar{C} \bar{\varphi}(t) &= \dot{\bar{V}}_{\bar{\varphi}}(t) + 2\alpha V_{\bar{\varphi}}(t) + \varphi(t)^\top C^\top C \varphi(t) \\ &= -2(\zeta - \alpha) \bar{\varphi}(t)^\top \bar{P}_i(t) \bar{\varphi}(t) + 2\bar{\varphi}(t)^\top \bar{P}_i(t) \bar{L}_i \bar{D}_i \bar{e}(t) \\ &\quad - 2\bar{\varphi}(t)^\top \bar{P}_i(t) \bar{L}_i \bar{D}_i \bar{\delta}_e(t) \end{aligned} \quad (63)$$

and

$$\begin{aligned} \dot{V}_e(t) + 2\alpha V_e(t) &= e(t)^\top \Phi_i(\alpha, Q) e(t) \\ &\quad + 2e(t)^\top QL_i D_i \delta_e(t) + 2e(t)^\top Q \chi_i(t) \\ &\quad + h^2 e^{2\alpha h} \dot{e}(t)^\top W \dot{e}(t) - \frac{\pi^2}{4} \delta_e(t)^\top W \delta_e(t), \end{aligned} \quad (64)$$

where we used the notation (29). Recalling (59), the upper bound (30) is replaced by

$$\bar{\Psi}_i(t) := \begin{bmatrix} -2(\zeta - \alpha)\bar{P}_i(t) + \gamma\kappa^2\Omega_1 & \bar{P}_i(t)\bar{L}_i D_i + \gamma\kappa^2\Omega_2 & -\bar{P}_i(t)\bar{L}_i D_i & 0 \\ * & \Phi_i(\alpha, Q) + \gamma\kappa^2 I_n + h^2 e^{2\alpha h} U_i^\top W U_i & Q L_i D_i + h^2 e^{2\alpha h} U_i^\top W L_i D_i & Q + h^2 e^{2\alpha h} U_i^\top W \\ * & * & -\frac{\pi^2}{4} W + h^2 e^{2\alpha h} D_i^\top L_i^\top W L_i D_i & h^2 e^{2\alpha h} D_i^\top L_i^\top W \\ * & * & * & -\gamma I_n + h^2 e^{2\alpha h} W \end{bmatrix}, \quad (59)$$

$$\Omega_1 = \begin{bmatrix} I_n & 0 \\ 0 & 0 \end{bmatrix} \in \mathbb{R}^{(n+1) \times (n+1)}, \quad \Omega_2 = \begin{bmatrix} I_n \\ 0 \end{bmatrix} \in \mathbb{R}^{(n+1) \times n}$$

$$\bar{N}_i(t) = \kappa^2 \begin{bmatrix} \bar{\varphi}(t) \\ e(t) \end{bmatrix}^\top \begin{bmatrix} \Omega_1 & \Omega_2 \\ * & I_n \end{bmatrix} \begin{bmatrix} \bar{\varphi}(t) \\ e(t) \end{bmatrix} - \|\chi_i(t)\|^2 \geq 0, \quad (65)$$

whereas (31) is replaced by

$$\dot{e}(t)^\top W \dot{e}(t) = \begin{bmatrix} e(t) \\ \delta_e(t) \\ \chi_i(t) \end{bmatrix}^\top \begin{bmatrix} U_i^\top \\ D_i^\top L_i^\top \\ I_n \end{bmatrix} W \begin{bmatrix} U_i^\top \\ D_i^\top L_i^\top \\ I_n \end{bmatrix} \begin{bmatrix} e(t) \\ \delta_e(t) \\ \chi_i(t) \end{bmatrix}. \quad (66)$$

Denoting

$$\bar{\xi}(t) = \text{col}\{\bar{\varphi}(t), e(t), \delta_e(t), \chi_i(t)\} \quad (67)$$

and letting $\gamma > 0$, a.e. in $[t_k, t_k + T)$, we have

$$\begin{aligned} & \dot{V}(t) + 2\alpha V(t) + \varphi(t)^\top C^\top C \varphi(t) \\ & \leq \dot{V}(t) + 2\alpha V(t) + \varphi(t)^\top C^\top C \varphi(t) + \gamma \bar{N}_i(t) \\ & = \bar{\xi}(t)^\top \bar{\Psi}_i(t) \bar{\xi}(t) \leq 0. \end{aligned}$$

Considering now the interval $[t_k + T, t_{k+1})$, where $\sigma(t) = i$ by (57) and the switching condition has not occurred, by arguments similar to (34)–(35), we have

$$\begin{aligned} & \dot{V}_{\bar{\varphi}}(t) + 2\alpha V_{\bar{\varphi}}(t) + \varphi(t)^\top C^\top C \varphi(t) \\ & = -2(\zeta - \alpha)\bar{\varphi}(t)^\top \bar{X}_i \bar{\varphi}(t) + 2\bar{\varphi}(t)^\top \bar{X}_i \bar{L}_i \bar{D}_i \bar{e}(t) \\ & \quad - 2\bar{\varphi}(t)^\top \bar{X}_i \bar{L}_i \bar{D}_i \bar{\delta}_e(t) + \varepsilon \bar{\varphi}(t)^\top E \bar{\varphi}(t) \\ & = -2(\zeta - \alpha)\bar{\varphi}(t)^\top \bar{X}_i \bar{\varphi}(t) + 2\bar{\varphi}(t)^\top \bar{X}_i \bar{L}_i \bar{D}_i \bar{e}(t) \\ & \quad - 2\bar{\varphi}(t)^\top \bar{X}_i \bar{L}_i \bar{D}_i \bar{\delta}_e(t) + \varepsilon, \end{aligned} \quad (68)$$

where we used the fact that $\bar{\varphi}(t)^\top E \bar{\varphi}(t) = 1$, in view of (52) and (55). Letting $\gamma > 0$, a.e. in $[t_k + T, t_{k+1})$, we have

$$\begin{aligned} & \dot{V}(t) + 2\alpha V(t) + \varphi(t)^\top C^\top C \varphi(t) \\ & \leq \dot{V}(t) + 2\alpha V(t) + \varphi(t)^\top C^\top C \varphi(t) + \gamma \bar{N}_i(t) \\ & = \bar{\xi}(t)^\top \bar{\Psi}_i(t) \bar{\xi}(t) + \varepsilon \leq \varepsilon. \end{aligned}$$

Summarising, we have that

$$\dot{V}(t) + 2\alpha V(t) + \varphi(t)^\top C^\top C \varphi(t) \leq \varepsilon, \quad \text{a.e. in } [t_k, t_{k+1}), \quad (69)$$

and the functional $V(t)$ exhibits no upward jumps at $\{t_k\}_{k=1}^\infty$. Employing arguments similar to the proof of Theorem 1, (69) yields (60), where $\mathcal{C}_1$ and $\mathcal{C}_3$ can be computed as in (24) and (25). The estimate (20) with $\bar{B}$ replaced by $\bar{B}_1$, which is given in (60), follows exactly as in Theorem 1. ■

4. LMI FEASIBILITY GUARANTEES AND DISCRETISATION

In this section, we study the feasibility of the matrix inequalities obtained in Section 3.2. For simplicity of presentation, we will focus on Theorem 1 and the feasibility of $\Psi_i(t) \prec 0$, $i \in [\ell]$, $t \in [0, T]$, defined in (16). However, we stress that similar arguments can be applied to the design conditions presented in Section 3.3 and Section 3.4.

Proposition 1. Under Assumptions 1 and 2, consider $P_i(t)$, $i \in [\ell]$, as in Theorem 1. Let $Q \in \mathbb{R}^{n \times n}$ with $Q \succ 0$ be such that, for all $i \in [\ell]$,

$$\Gamma_i(t) := \begin{bmatrix} -2\zeta P_i(t) & P_i(t) L_i D_i \\ * & \Phi_i(0, Q) \end{bmatrix} \prec 0, \quad t \in [0, T]. \quad (70)$$

Then, $\Psi_i(t) \prec 0$, $t \in [0, T]$, holds for all $i \in [\ell]$, provided that $h, \alpha, \kappa > 0$, are small enough and $\gamma > 0$ is large enough.

Proof. Consider (16) and set $\alpha = 0$, $h = \kappa = \gamma^{-\beta}$, with $\beta > \frac{1}{2}$, and $W = \gamma I_n$. Then, the bottom-right 2×2 block in (16) has the form

$$-\gamma \left(\begin{bmatrix} \frac{\pi^2}{4} I_n & 0 \\ 0 & I_n \end{bmatrix} - \gamma^{-2\beta} \begin{bmatrix} D_i^\top L_i^\top L_i D_i & D_i^\top L_i^\top \\ * & I_n \end{bmatrix} \right) =: -\gamma \Xi(\gamma).$$

Since $\lim_{\gamma \rightarrow \infty} \Xi(\gamma) = \begin{bmatrix} \frac{\pi^2}{4} I_n & 0 \\ 0 & I_n \end{bmatrix} \succ 0$, we have $-\gamma \Xi(\gamma) \prec 0$ for $\gamma \geq \gamma_* > 0$, with γ_* large enough, whence $h, \kappa > 0$ small enough. Then, applying Schur complement in (16), with respect to this 2×2 block, we have that $\Psi_i(t) \prec 0$, $t \in [0, T]$, holds if and only if

$$\begin{aligned} & \Gamma_i(t) + \begin{bmatrix} \gamma^{1-2\beta} I_n & \gamma^{1-2\beta} I_n \\ * & \gamma^{1-2\beta} (I_n + U_i^\top U_i) \end{bmatrix} \\ & + \gamma^{-1} \Delta_i(\gamma, t) \Xi(\gamma)^{-1} \Delta_i(\gamma, t)^\top \prec 0, \quad t \in [0, T], \end{aligned} \quad (71)$$

where we note that $1 - 2\beta < 0$ and define

$$\Delta_i(\gamma, t) := \begin{bmatrix} -P_i(t) L_i D_i & 0 \\ Q L_i D_i + \gamma^{1-2\beta} U_i^\top L_i D_i & Q + \gamma^{1-2\beta} U_i^\top U_i \end{bmatrix}.$$

Since P_i is continuous on $[0, T]$, so are $\Gamma_i(t)$ and $\Delta_i(\gamma, t)$. Thus, in view of compactness of $[0, T]$, there exist some $\mu, \eta > 0$ such that, for all $t \in [0, T]$, we have $\Gamma_i(t) \prec -\mu I_n$ and $\sup_{t \in [0, T], \gamma \geq \gamma_*} \|\Delta_i(\gamma, t)\| \leq \eta$, for all $i \in [\ell]$. Thus, by increasing γ_* if necessary, (71) can be guaranteed to hold. Taking again into account that $P_i(t)$, $i \in [\ell]$, are continuous on $[0, T]$, similar arguments show there exists some $\alpha_* > 0$ such that feasibility of $\Psi_i(t) \prec 0$, $i \in [\ell]$ for all $t \in [0, T]$ is guaranteed for $\alpha \in (0, \alpha_*)$. ■

Proposition 2. Assume that $\Psi_i(t) \prec 0$, $t \in [0, T]$, for all $i \in [\ell]$, with $\Psi_i(t)$ given in (16). Then, (70) holds for all $i \in [\ell]$ with the same $Q \in \mathbb{R}^{n \times n}$ and $\zeta > 0$ as in (16).

Proof. For $i \in [\ell]$, $\Psi_i(t) \prec 0$, $t \in [0, T]$, implies that

$$\begin{aligned} 0 & \succ \begin{bmatrix} -2(\zeta - \alpha)P_i(t) + \gamma\kappa^2 I_n & P_i(t) L_i D_i + \gamma\kappa^2 I_n \\ * & \Phi_i(\alpha, Q) + \gamma\kappa^2 I_n + h^2 e^{2\alpha h} U_i^\top W U_i \end{bmatrix} \\ & = \Gamma_i(t) + \begin{bmatrix} 2\alpha P_i(t) & 0 \\ 0 & 2\alpha Q + h^2 e^{2\alpha h} U_i^\top W U_i \end{bmatrix} + \gamma\kappa^2 \begin{bmatrix} I_n & I_n \\ I_n & I_n \end{bmatrix} \\ & =: \Gamma_i(t) + M(t) + M_0, \quad t \in [0, T]. \end{aligned} \quad (72)$$

Since $\alpha, h > 0$ and $Q, W, P_i(t) \succ 0$, $t \in [0, T]$, we have $M(t) \succ 0$, $t \in [0, T]$. Also, since $\gamma, \kappa > 0$, it follows that $M_0 \succ 0$; see (30). Therefore

$$\Gamma_i(t) \prec \Gamma_i(t) + M(t) + M_0 \prec 0,$$

and this concludes the proof. ■

Propositions 1 and 2 together provide a sufficient and necessary condition for the feasibility of $\Psi_i(t) \prec 0$ for *some* tuning parameters, summarised in the following theorem.

Theorem 3. $\Psi_i(t) \prec 0$, $t \in [0, T]$, holds $\forall i \in [\ell]$ for some $\alpha > 0$, inter-sampling time $h > 0$ and Lipschitz constants $\kappa > 0$, if and only if (70) holds $\forall i \in [\ell]$.

To verify the feasibility of $\Psi_i(t) \prec 0$, $t \in [0, T]$, with $\Psi_i(t)$ given in (16), we propose to discretise the interval $[0, T]$ into a densely-spaced grid $0 = \tau_0 < \dots < \tau_M = T$ and simultaneously verify the LMIs $\Psi_i(\tau_j) \prec 0$, $i \in [\ell]$, $j \in [M]$. The following results formally justify such a discretisation.

Lemma 1. Let $X_i \in \mathbb{R}^{n \times n}$ with $X_i \succ 0$ be fixed and consider $P_i(t)$ satisfying (15). Then,

$$\max_{t \in [0, T]} \|P_i(t)\| \leq (\|X_i\| + T \|C^\top C\|) e^{2T\|A_{i,\zeta}\|} =: \mathbf{P}_i,$$

where $A_{i,\zeta} := A_i + \zeta I_n$.

Proof. For $\Sigma_i(t) := P_i(T - t)$, $t \in [0, T]$, in view of (15), $\dot{\Sigma}_i = A_{i,\zeta}^\top \Sigma_i + \Sigma_i A_{i,\zeta} + C^\top C$. Integrating on $[0, \tau]$ with $0 \leq \tau \leq T$, since $\Sigma_i(0) = X_i$, yields

$$\begin{aligned} \Sigma_i(\tau) &= X_i + \tau C^\top C \\ &+ A_{i,\zeta}^\top \int_0^\tau \Sigma_i(s) ds + \int_0^\tau \Sigma_i(s) ds A_{i,\zeta}, \quad \tau \in [0, T]. \end{aligned} \quad (73)$$

Since $\max_{t \in [0, T]} \|P_i(t)\| = \max_{\tau \in [0, T]} \|\Sigma_i(\tau)\|$, the result follows from taking the norm on both sides of (73), using norm sub-multiplicativity, exchanging the norm with the integrals, and employing the Grönwall inequality. ■

Lemma 2. Let $X_i \in \mathbb{R}^{n \times n}$ with $X_i \succ 0$ be fixed and consider $P_i(t)$ satisfying (15). Given $\varepsilon > 0$, let $\mu > 0$ be such that

$$\begin{aligned} \vartheta_i(\mu) &:= \mu e^{2\|A_{i,\zeta}\|\mu} \|C^\top C\| \\ &+ \left[2 \left(e^{\|A_{i,\zeta}\|\mu} - 1 \right) + \left(e^{\|A_{i,\zeta}\|\mu} - 1 \right)^2 \right] \mathbf{P}_i < \varepsilon. \end{aligned}$$

Then, for any $0 \leq \tau_0 \leq \tau_1 \leq T$ with $\tau_1 - \tau_0 < \mu$, it holds $\|P_i(\tau_1) - P_i(\tau_0)\| < \varepsilon$.

Proof. Employing (15a), we have

$$0 = \frac{d}{dt} \left(e^{A_{i,\zeta}^\top t} P_i(t) e^{A_{i,\zeta} t} \right) + e^{A_{i,\zeta}^\top t} C^\top C e^{A_{i,\zeta} t}.$$

Integrating the latter equality on $[\tau_0, \tau_1]$, employing some algebraic manipulations and denoting $\Delta\tau := \tau_1 - \tau_0$ and $\mathcal{I} := \int_0^{\Delta\tau} e^{A_{i,\zeta}^\top s} C^\top C e^{A_{i,\zeta} s} ds$, we end up with

$$\begin{aligned} P_i(\tau_0) - P_i(\tau_1) &= \mathcal{I} + e^{A_{i,\zeta}^\top \Delta\tau} P(\tau_1) e^{A_{i,\zeta} \Delta\tau} - P(\tau_1) \\ &= \mathcal{I} + P_i(\tau_1) h_i^\zeta(\Delta\tau) + h_i^\zeta(\Delta\tau)^\top P_i(\tau_1) \\ &\quad + h_i^\zeta(\Delta\tau)^\top P_i(\tau_1) h_i^\zeta(\Delta\tau), \end{aligned}$$

where $h_i^\zeta(\mu) := e^{A_{i,\zeta}\mu} - I_n$ satisfies $\|h_i^\zeta(\mu)\| \leq e^{\|A_{i,\zeta}\|\mu} - 1$. Taking the norm of $P_i(\tau_1) - P_i(\tau_0)$ and using Lemma 1, we obtain

$$\|P_i(\tau_1) - P_i(\tau_0)\| \leq \vartheta_i(\Delta\tau) \leq \vartheta_i(\mu) < \varepsilon. \quad \blacksquare$$

Corollary 1. Consider (16) with fixed scalars $\zeta, h, \gamma, \kappa > 0$ and $\alpha \in [0, \zeta)$, and fixed matrices $W, Q \in \mathbb{R}^{n \times n}$, $W, Q \succ 0$. Under the assumptions of Lemma 2, let $\mu > 0$ be such that $\vartheta_i(\mu) < \frac{\varepsilon}{2(\zeta - \alpha + \|L_i D_i\|)}$. Then,

$$\|\Psi_i(\tau_1) - \Psi_i(\tau_0)\| < \varepsilon.$$

Proof. Employing (16) and Lemma 2, we have

$$\begin{aligned} \|\Psi_i(\tau_1) - \Psi_i(\tau_0)\| &\leq 2(\zeta - \alpha + \|L_i D_i\|) \|P_i(\tau_1) - P_i(\tau_0)\| \\ &\leq 2(\zeta - \alpha + \|L_i D_i\|) \vartheta_i(\Delta\tau) \\ &\leq 2(\zeta - \alpha + \|L_i D_i\|) \vartheta_i(\mu) < \varepsilon. \quad \blacksquare \end{aligned}$$

We can now formulate the following theorem.

Theorem 4. Let $i \in [\ell]$ be fixed and consider the grid $0 = \tau_0 < \dots < \tau_M = T$ on $[0, T]$. Given $\varepsilon, \zeta > 0$ and $\alpha \in (0, \zeta)$, assume that

$$\mathfrak{s} := \max_{0 \leq j \leq M-1} (\tau_{j+1} - \tau_j) < \mu, \quad (74)$$

with $\mu > 0$ such that $\vartheta_i(\mu) < \frac{\varepsilon}{2(\zeta - \alpha + \|L_i D_i\|)}$. Consider $h, \gamma, \kappa > 0$ and $W, Q \in \mathbb{R}^{n \times n}$ with $W, Q \succ 0$ such that, for all $k \in [M]$, the LMI $\Psi_i(\tau_k) \prec -\varepsilon I_n$ holds. Then, $\Psi_i(t) \prec 0$ for all $t \in [0, T]$.

Proof. For $t \in [0, T]$, by assumption, there exists $j \in [M]$ such that $0 < \tau_j - t \leq \mathfrak{s} < \mu$. Employing Corollary 1, we conclude that $\|\Psi_i(t) - \Psi_i(\tau_j)\| < \varepsilon$. By Weyl's inequality (Horn and Johnson, 2012, Section 4.3), the corresponding maximal eigenvalues satisfy

$$|\lambda_{\max}(\Psi_i(t)) - \lambda_{\max}(\Psi_i(\tau_j))| < \varepsilon.$$

Hence, $\lambda_{\max}(\Psi_i(t)) < \lambda_{\max}(\Psi_i(\tau_j)) + \varepsilon < 0$, implying that $\Psi_i(t) \prec 0$. Since $t \in [0, T]$ is arbitrary, the proof is concluded. ■

Remark 10. Theorem 4 provides a sufficient condition for guaranteeing the feasibility of the LMI $\Psi_i(t) \prec 0$, $t \in [0, T]$ via discretisation. However, in practice, this method will likely yield a conservative bound on the stencil size $\mathfrak{s}$ in (74); tighter estimates on $\mathfrak{s}$ should be attainable via direct computation in concrete examples.

5. NUMERICAL EXAMPLES

We validate the efficacy of the proposed observer-based sampled-data switching law with dwell-time via two numerical examples, which demonstrate that our approach can stabilise switched systems when all the individual modes have unstable matrices A_i , and even when the matrices A_i do not admit a Hurwitz convex combination. The proposed switching strategy is also applied to control a double buck-boost converter, confirming its practical effectiveness in real-world applications to power electronics.

Throughout the examples, given system (1), we choose observer gains by solving the simplified LMIs in (50) and setting $L_i = Q^{-1} Y_i$, $i \in [\ell]$, so that Assumption 2 is satisfied. For a chosen $\Pi \in \mathcal{M}_n^1$ and $T, \zeta > 0$, we find X_i that satisfy (12), based on which we then compute $P_i(t)$ by integrating (15). Then, treating $h, \alpha > 0$ as tuning parameters, we check the existence of $Q, W \succ 0$ and $\gamma > 0$ such that $\Psi_i(t) \prec 0$ is satisfied $\forall i \in [\ell]$ and $\forall t \in [0, T]$ by verifying the LMIs on a uniform grid with stencil $\mathfrak{s} = 0.01$, where $\mathfrak{s}$ is defined in (74). We verify that the switching law (14) ensures global asymptotic stabilisation by simulating the closed-loop system.

Example 1. (Unstable nominal modes.) Consider system (1) with two modes, $\sigma(t) \in \{1, 2\}$, matrices

$$A_1 = \begin{bmatrix} -2 & 0.3 \\ -2 & 1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 1 & 2 \\ -0.3 & -4 \end{bmatrix},$$

¹ The selection of $\Pi \in \mathcal{M}_n$ depends on whether a stabilising convex combination exists: if so, Π is chosen such that $\lambda^\top \Pi = 0$, where λ is the vector of convex coefficients; otherwise, Π is interpreted as the infinitesimal generator of a Markov process where $\Pi_{i,j}$ denotes the expected transition rate from subsystem i to j (see Russo et al. (2025) and references therein).

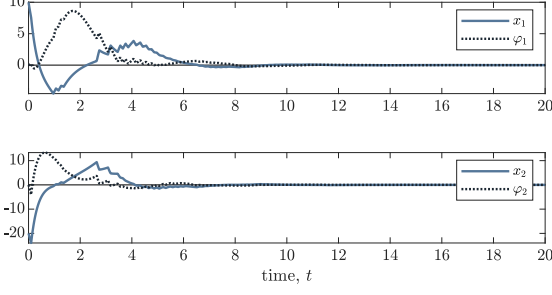


Fig. 1. Time evolution of the state x (blue) and of the observed state φ (black) in Example 1.

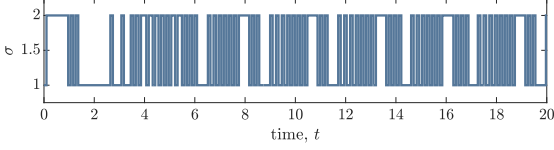


Fig. 2. Time evolution of the switching law $\sigma(t)$ in Example 1.

$$D_1 = D_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad C = 10^{-5} \cdot \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

and functions $f_i(x) = \kappa \frac{\|x\|}{1+\|x\|} [1 \ 1]^\top$, with $\kappa = 0.002$. Both A_1 and A_2 are unstable. However, their convex combination $A_\lambda = \lambda A_1 + (1 - \lambda) A_2$, with $\lambda = 0.5$, is a Hurwitz matrix. We choose observer gains

$$L_1 = \begin{bmatrix} -1.50 & -0.85 \\ -0.85 & 1.50 \end{bmatrix}, \quad L_2 = \begin{bmatrix} 1.50 & 0.85 \\ 0.85 & -3.50 \end{bmatrix}$$

that satisfy Assumption 2 and select matrix Π in (12) as

$$\Pi = \begin{bmatrix} -21.21 & 21.21 \\ 21.21 & -21.21 \end{bmatrix}.$$

The simulation results for the closed-loop system with the switching law (14), inter-sampling interval $h = 0.05$, dwell time $T = 0.1$, parameters $\zeta = 0.1$ and $\alpha = 10^{-6}$, are shown in Fig. 1. The closed-loop system trajectories converge to the origin, despite the instability of the two nominal modes and the unknown nonlinear term. The switching law (14), shown in Fig. 2, satisfies the dwell-time constraint.

Example 2. (Nominal modes with no Hurwitz convex combination.)

Consider a switched system as in (1) with three modes, $\sigma(t) \in \{1, 2, 3\}$, matrices

$$A_1 = \begin{bmatrix} -\eta & 0 & 0 \\ 0 & 0 & 0 \\ 0 & \beta & 0 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0 & 0 & \beta \\ 0 & -\eta & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad A_3 = \begin{bmatrix} 0 & 0 & 0 \\ \beta & 0 & 0 \\ 0 & 0 & -\eta \end{bmatrix},$$

$$D_1 = D_2 = D_3 = [1 \ 1 \ 1], \quad C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

and functions $f_i(x) = \kappa [\sin(x_1) \sin(x_2) \sin(x_3)]^\top$, with $\kappa = 0.002$. Matrices A_i are taken from a well-known congestion example (see, e.g., Blanchini et al., 2012) and are characterised by the absence of a Hurwitz convex combination. Hence, switching approaches such as the ones by Bolzern and Spinelli (2004); Albea et al. (2019); Albea and Seuret (2021) cannot be applied to stabilise the system. We select the system parameters as $\eta = 1$ and $\beta = 1.1$

We choose observer gains

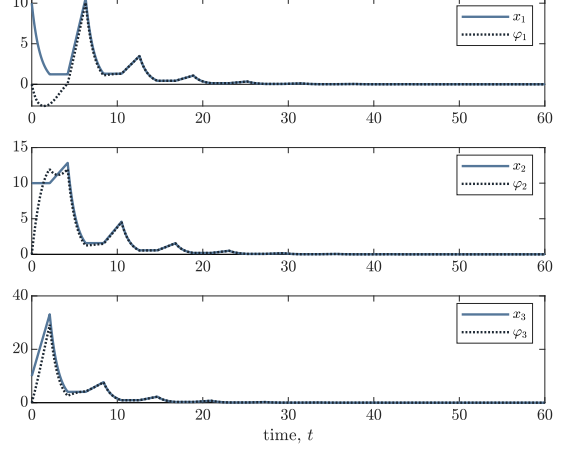


Fig. 3. Time evolution of the state x (blue) and of the observed state φ (black) in Example 2.

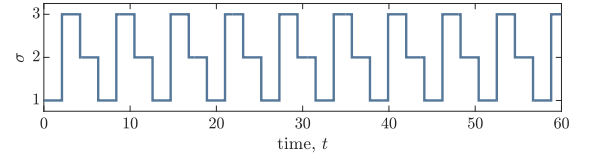


Fig. 4. Time evolution of the switching law $\sigma(t)$ in Example 2.

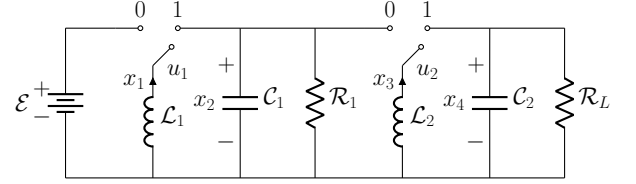


Fig. 5. Topology of a double buck-boost converter.

$$L_1 = \begin{bmatrix} -0.51 \\ 0.53 \\ 0.53 \end{bmatrix}, \quad L_2 = \begin{bmatrix} 0.53 \\ -0.51 \\ 0.53 \end{bmatrix},$$

that satisfy Assumption 2 and select matrix Π in (12) as

$$\Pi = \begin{bmatrix} -10 & 0 & 10 \\ 10 & -10 & 0 \\ 0 & 10 & -10 \end{bmatrix}.$$

The simulation results for the closed-loop system with the switching law in Theorem 1, inter-sampling interval $h = 0.2$, dwell time $T = 2.1$, parameters $\zeta = 0.1$ and $\alpha = 10^{-6}$, are shown in Fig. 3, while the switching law is shown in Fig. 4. The switching law (14) stabilises the closed-loop system and satisfies the dwell-time constraint.

Example 3. (Double buck-boost converter.)

Double buck-boost converters are switching power electronic devices recognised for their ability to independently manage two loads with distinct steady-state or tracking requirements (Sira-Ramirez and Silva-Ortigoza, 2006, Section 2.11). As illustrated in Fig. 5, the topology incorporates two switching elements, yielding four distinct operating configurations. Consequently, the converter can be modelled as the switched affine system (1) with $\theta = 1$, $f_i \equiv 0$, $\sigma(t) \in [4]$, and

$$A_1 = \begin{bmatrix} 0 & \mathcal{L}_1^{-1} & 0 & 0 \\ -C_1^{-1} - (\mathcal{R}_1 C_1)^{-1} & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathcal{L}_2^{-1} \\ 0 & 0 & -C_2^{-1} - (\mathcal{R}_L C_2)^{-1} \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix},$$

$$\begin{aligned}
A_2 &= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & -(\mathcal{R}_1 \mathcal{C}_1)^{-1} & 0 & 0 \\ 0 & 0 & 0 & \mathcal{L}_2^{-1} \\ 0 & 0 & -\mathcal{C}_2^{-1} & -(\mathcal{R}_L \mathcal{C}_2)^{-1} \end{bmatrix}, \quad b_2 = \begin{bmatrix} \mathcal{E} \mathcal{L}_1^{-1} \\ 0 \\ 0 \\ 0 \end{bmatrix}, \\
A_3 &= \begin{bmatrix} 0 & \mathcal{L}_1^{-1} & 0 & 0 \\ -\mathcal{C}_1^{-1} & -(\mathcal{R}_1 \mathcal{C}_1)^{-1} & -\mathcal{C}_1^{-1} & 0 \\ 0 & \mathcal{L}_2^{-1} & 0 & 0 \\ 0 & 0 & 0 & -(\mathcal{R}_L \mathcal{C}_2)^{-1} \end{bmatrix}, \quad b_3 = b_1, \\
A_4 &= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & -(\mathcal{R}_1 \mathcal{C}_1)^{-1} & -\mathcal{C}_1^{-1} & 0 \\ 0 & \mathcal{L}_2^{-1} & 0 & 0 \\ 0 & 0 & 0 & -(\mathcal{R}_L \mathcal{C}_2)^{-1} \end{bmatrix}, \quad b_4 = b_2,
\end{aligned}$$

In this formulation, the state variables are defined as follows: x_1 and x_3 represent the currents through inductors $\mathcal{L}_1$ and $\mathcal{L}_2$, respectively. The voltage variables x_2 and x_4 correspond to the capacitor voltages at each conversion stage; specifically, x_2 is the voltage across $\mathcal{C}_1$ at the first stage, while x_4 represents the voltage across $\mathcal{C}_2$, which serves as the final output or load voltage. The switching signal $\sigma(t)$ denotes the configurations of the pair of switches (u_1, u_2) in Fig. 5, with $\sigma(t) = 1$ corresponding to the configuration $(0, 0)$, $\sigma(t) = 2$ to $(1, 0)$, $\sigma(t) = 3$ to $(0, 1)$ and, finally, $\sigma(t) = 4$ to $(1, 1)$. The objective of the double buck-boost converter is to regulate the voltages at the two output stages, specifically x_2 and x_4 , to their respective reference values, x_2^* and x_4^* . Hence, for given x_2^* and x_4^* , a parametrisation of the equilibrium point in terms of the steady-state output voltages is obtained as

$$x_1^* = \left(\frac{x_2^*}{\mathcal{R}_1} + \frac{x_4^*}{x_2^* \mathcal{R}_L} \right) \frac{x_2^* - \mathcal{E}}{\mathcal{E}}, \quad x_3^* = \frac{x_4^* (x_4^* - x_2^*)}{\mathcal{R}_L x_2^*}.$$

Therefore, the control objective is to regulate to zero the error variable $x^e = x - x^*$, $x^* = [x_1^* \ x_2^* \ x_3^* \ x_4^*]^\top$, obeying the switched system

$$\begin{aligned}
\dot{x}^e(t) &= A_{\sigma(t)} x^e(t) + b_{\sigma(t)}^e, \\
y^e(t) &= D_{\sigma(t)} x^e(s_k), \\
z^e(t) &= C x^e(t),
\end{aligned} \tag{75}$$

where $b_{\sigma(t)}^e = A_{\sigma(t)} x^* + b_{\sigma(t)}$. The numerical values for the converter parameters are selected as in (Sira-Ramirez and Silva-Ortigoza, 2006, Section 2.10). Given the input voltage $\mathcal{E} = 12V$, with $x_2^* = -12V$ and $x_4^* = 12V$, the convex combination of $\{A_i\}_{i=1}^4$ obtained with coefficients $\lambda_1 = \lambda_4 = 0.5$, $\lambda_2 = \lambda_3 = 0$, results in a stable matrix. We consider our system with $D_i = I_4$, $i \in [4]$, which allows for comparison with Hetel and Fridman (2013), and we choose the observer gains

$$\begin{aligned}
L_1 &= 10^{-6} \cdot \begin{bmatrix} 0.500 & -0.042 & 0.002 & -0.019 \\ -0.042 & 160.105 & -0.038 & -0.016 \\ 0.002 & -0.038 & 0.204 & -0.301 \\ -0.019 & -0.016 & -0.301 & 64.623 \end{bmatrix}, \\
L_2 &= 10^{-6} \cdot \begin{bmatrix} 0.750 & -0.064 & 0.003 & -0.028 \\ -0.064 & 240.158 & -0.058 & -0.024 \\ 0.003 & -0.058 & 0.307 & -0.452 \\ -0.028 & -0.024 & -0.452 & 96.935 \end{bmatrix}, \\
L_3 &= 10^{-6} \cdot \begin{bmatrix} 1.000 & -0.085 & 0.004 & -0.038 \\ -0.085 & 320.211 & -0.077 & -0.032 \\ 0.004 & -0.077 & 0.409 & -0.603 \\ -0.038 & -0.032 & -0.603 & 129.247 \end{bmatrix}, \\
L_4 &= 10^{-6} \cdot \begin{bmatrix} 1.250 & -0.106 & 0.005 & -0.047 \\ -0.106 & 400.263 & -0.096 & -0.040 \\ 0.005 & -0.096 & 0.511 & -0.753 \\ -0.047 & -0.040 & -0.753 & 161.558 \end{bmatrix},
\end{aligned}$$

that satisfy Assumption 2, as well as the matrix Π in (12) as

$$\Pi = 10^8 \cdot \begin{bmatrix} -0.7071 & 0 & 0 & 0.7071 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \\ 0.7071 & 0 & 0 & -0.7071 \end{bmatrix}.$$

Finally, we set $\zeta = 10^{-3}$ and $\alpha = 10^{-4}$. Due to the high switching frequency required for DC-DC power converters, both the dwell time and the maximum sampling interval are set to $T = 10^{-5}$ s and $h = 10^{-5}$ s, while the converter is assumed to be initially discharged, i.e., $x(0) = 0$. In this example, it is assumed that the affine vectors b_i are known, and it is hence possible to implement the switching strategy outlined in Section 3.4 and summarised in Theorem 2. The proposed switching law is compared with the algorithm proposed by Hetel and Fridman (2013), which addresses the state feedback stabilisation of switched affine systems under sampled-data control. To ensure a fair comparison between the two approaches, the sampling interval for the algorithm by Hetel and Fridman (2013) is set to h , matching the sampling interval length used in our proposed approach. The performance of our proposed switching law is shown in Fig. 6, demonstrating that the states converge towards the desired reference values and oscillate around them. The obtained performance is comparable to the performance obtained from the approach by Hetel and Fridman (2013), in terms of both transient and steady-state dynamics. In fact, for the two capacitor voltages x_2 and x_4 , our switching strategy maintains a steady-state relative error below 0.76% and 0.18%, respectively. In comparison, the method proposed by Hetel and Fridman (2013) yields steady-state relative errors below 0.61% and 0.10% for x_2 and x_4 , respectively. Furthermore, the proposed algorithm yields a steady-state RMS error equal to 0.047 V and 0.010 V for x_2 and x_4 , respectively, while the algorithm presented in Hetel and Fridman (2013) guarantees a steady-state RMS error equal to 0.042 V and 0.006 V, for x_2 and x_4 , respectively. As shown in Fig. 7, the switching signals obtained by both algorithms eventually converge to a repeated switching between modes 1 and 4.

While the design of the switching strategy by Hetel and Fridman (2013) imposes a dwell-time constraint that is equal to the sampling period, our proposed methodology decouples the dwell time from the sampling time. This separation allows multiple switching events within a single sampling interval, a feature particularly useful when the sampling and switching are not collocated and the measurements are acquired at a lower frequency than the maximum achievable switching frequency. This advantage is clearly demonstrated in Fig. 8, where both algorithms are implemented with a sampling period of 10^{-5} s, while our approach employs a shorter dwell time of 10^{-6} s. As illustrated in the figure, our proposed algorithm achieves significantly lower voltage oscillation amplitudes around the desired set-points, due to the refined dwell-time control. In fact, in this case our approach yields a steady-state relative error below 0.067% and 0.012% for the two capacitor voltages, respectively. Furthermore, adopting smaller dwell time guarantees a steady-state RMS error equal to 0.0046 V and 0.0007 V, for x_2 and x_4 , respectively.

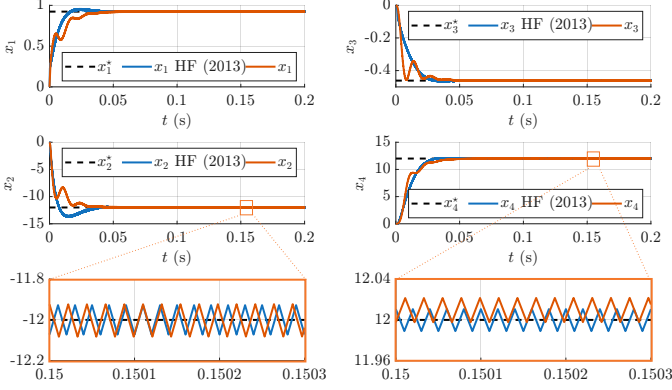


Fig. 6. Time-evolution of the double buck-boost converter states obtained with our proposed switching strategy (red solid line) and the strategy by Hetel and Fridman (2013) (blue solid line); the black dashed line is the reference signal. Top left: current $x_1(t)$. Bottom left: first stage voltage $x_2(t)$. Top right: current $x_3(t)$. Bottom right: second stage voltage $x_4(t)$.

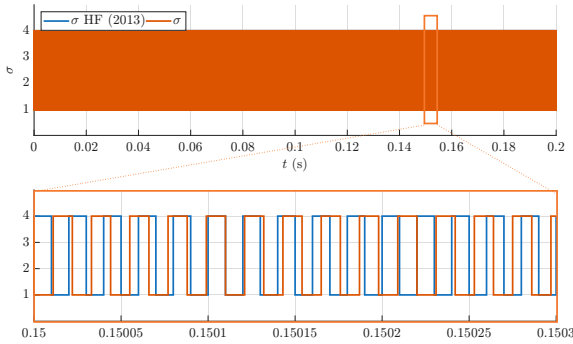


Fig. 7. Time-evolution of the double buck-boost converter switching signal obtained with the proposed switching strategy (red solid line) and the strategy in Hetel and Fridman (2013) (blue solid line).

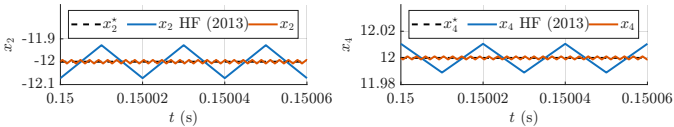


Fig. 8. Ultimate bound of the double buck-boost converter voltages obtained with our proposed switching strategy (red solid line) and the strategy by Hetel and Fridman (2013) (blue solid line); the black dashed line is the reference signal. In our approach, the dwell time is smaller than the sampling time, whereas the strategy by Hetel and Fridman (2013) forces the sampling period to be equal to the dwell-time constraint.

6. CONCLUDING DISCUSSION

We proposed a novel observer-based switching law with dwell-time constraints to achieve either stability or ultimate boundedness of switched systems with nominal linear-affine dynamics, subject to uncertain Lipschitz nonlinearities and sampled-data output measurements. Our switching law relies on a suitably designed observer and Lyapunov-Metzler inequalities. We performed Lyapunov stability analysis, leading to time-dependent LMI conditions that provide a lower bound on the maximum inter-

sampling time, and of the Lipschitz constants κ_i , for which the closed-loop system is globally asymptotically stable (or, in the presence of switching affine terms, ultimately bounded) with the designed switching law. We investigated the feasibility of the derived LMIs and (i) showed how they can be modified to yield observer gains design; (ii) provided equivalent reduced-order LMI conditions; and (iii) proved that the time dependence of the LMIs can be removed by discretisation on a finite grid. Numerical simulations confirmed our theoretical results. We also compared our method to an existing approach within the literature, by Hetel and Fridman (2013), which deals with sampled-data state-feedback stabilisation, without including observers and without allowing for dwell-time constraints independent of the inter-sampling time of the state.

Although, to the best of our knowledge, no other work addresses the same framework of switched affine systems with nonlinearities under dwell-time constraints, sampled-data measurements and output feedback, other contributions in the literature share similarities with specific aspects of our problem formulation, but with important differences. For instance, Perez and Mera (2015) consider the robust stabilisation of nonlinear switched systems affected by bounded perturbations and by a control input $u(t)$, and propose an observer-based switching controller using sampled and quantised output measurements. As in our case, the nonlinearities satisfy a quasi-Lipschitz condition. A dwell-time constraint is also considered, and sufficient conditions for practical stability are formulated through Bilinear Matrix Inequalities derived from an extended invariant ellipsoid method. However, the approach by Perez and Mera (2015) differs from ours in several key aspects: their switching law is time-dependent, with in-built avoidance of Zeno behaviour via the introduction of a dwell-time, whereas we design an observer-dependent switching signal (14); their practical stability result relies on the presence of a control input $u(t)$ in the dynamics (which relies on the designed observer), whereas in our case stabilisation/ultimate boundedness is achieved through the observer-based switching signal design only; their stability analysis employs a Lyapunov functional suitable for general time-delay systems, more conservative than the Wirtinger-based functional we use; they do not employ Lyapunov-Metzler inequalities in their design.

Li et al. (2021) consider interval observer-based controller design for switched linear (but not linear-affine) systems, with Lipschitz continuous disturbance and measurement noise, relying on the observer designed by (Ethabet et al., 2018), but neither of these works considers sampled-data measurements. Also, in both works, the switching signal is time-dependent, with the avoidance of Zeno behaviour imposed via an appropriate dwell time. Li et al. (2021) introduce a controller $u(t)$ into the dynamics to achieve stability under *average* dwell time constraints. Moreover, both works impose structural assumptions on the system measurements by requiring that there exist observer gains $\{L_i\}_{i \in [\ell]}$ such that $A_i - L_i C_i$ is Metzler for all $i \in [\ell]$, in order to apply techniques from cooperative systems.

Zhao et al. (2020) propose an observer-based reliable control framework for switched linear (but not linear-affine) systems; the measurements are performed in continuous time and the time-dependent switching mechanism is ex-

ogenous. An output-feedback event-triggered controller $u(t)$ is designed in order to achieve stability of the closed-loop system; sampling only appears due to the use of an event-triggering mechanism in the controller. The analysis relies on the separation principle for the observer and state dynamics, and cannot be applied in the presence of Lipschitz nonlinearities.

Analogously, Fu and Li (2021) consider in Chapter 3 linear (but not linear-affine) switched systems subject to continuous time measurements and an exogenously given time-dependent switching signal, and investigate the stabilisation of the system via an event-triggered observer-based controller $u(t)$; also here, sampling only appears due to the use of an event-triggering mechanism in the controller, and the absence of nonlinearities in the system again allows for separation of observer and controller designs.

Zhang et al. (2025) consider only observation of a controlled switched system, subject to exogenously given time-dependent switching signal, sampled-data state measurements and uncertain nonlinearities, with the control objective being the achievement of exponentially ultimately bounded estimating error. The unknown nonlinearity is subject to global Lipschitz conditions, with the Lipschitz bound defining the ultimate bound on the estimation error. The observer is constructed in Luenberger form and ultimate boundedness is shown via Lyapunov analysis leading to sum of squares and LMI conditions. However, they consider neither the stabilisation/ultimate boundedness of the original state equation nor the design of a state/output-based switching signal, and do not employ Lyapunov-Metzler inequalities.

Finally, Carneiro et al. (2024) investigate practical stabilisation of a continuous-time uncertain switched affine system, with a switching affine term, subject to continuous time measurements, corrupted by noise. They employ a Luenberger observer whose state is then sampled at discrete time instants and used for synthesis of the switching signal. The switching law design and the stability analysis in the presence of sampled-data by Carneiro et al. (2024) is similar to that by Hetel and Fridman (2013) (in both, the sampling instants coincide with the switching instants, differently from our framework where the dwell time and the sampling time are decoupled) and assumes the existence of a convex combination $A(\kappa)$ of the mode matrices $\{A_i\}$ that is Hurwitz, as well as the fact that the pairs $(A(\kappa), C_i)$, where C_i are the measurement operators, are all observable. In comparison, our approach, which also handles the presence of affine switching dependent terms in the dynamics, in addition to Lipschitz nonlinearities, does not require a Hurwitz convex combination $A(\kappa)$ to exist, as also demonstrated in Example 2, where the nominal modes do not admit a Hurwitz convex combination.

Future work includes further extending our approach to consider general time-varying delays and more general classes of switched systems.

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